Title: Electrical and Electronic Component Environmental Compatibility Test

A INDEX

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B SCOPE

This Test Procedure is applicable to electrical and electronic components used on Jaguar Land Rover vehicles.

C OBJECT

This document describes the test methods and equipment requirements to be used to verify that electrical and electronic modules (including sensors, actuators and switches) satisfy the component level environmental testing for EC-7128/STJLR.18.233 requirements.

Testing shall not commence without written approval from Jaguar Land Rover of a test plan.

D RISK ASSESSMENT

A valid risk assessment must be in place before these tests are conducted.

E REVIEW

This document must be reviewed within three years of the date of the last review or update.

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F RELATED DOCUMENTS

References cited at the head of a test indicate that the test in this document is either more stringent or equivalent to the reference test cited.

Table 1 - Related Documents

Ref.	Document	Title	Location
1	**EC-7128**	E/E System Environmental (NON-EMC)	JLR SDOTs/ RMDV DOORS
2	**STJLR.18.233**	Electrical and Electronic Component Environmental Requirements (Non EMC)	JLR Standards Website
3	**27-7011**	Heat Management - Component Thermal Robustness	JLR SDOTs/ RMDV DOORS
4	**STJLR.00.5056**	Wet and Dry Zone Definition	JLR Standards Website
5	**ISO 16750**	Road vehicles — Environmental conditions and testing for electrical and electronic equipment	ISO Website
6	**ISO 20653**	Road vehicles — Degrees of protection (IP code) — Protection of electrical equipment against foreign objects, water and access	ISO Website
7	**IEC 60068**	Environmental testing	IEC Website
8	**ISO 12103-1**	Road vehicles — Test contaminants for filter evaluation — Part 1	ISO Website
9	**MIL-STD-810E**	Environmental Engineering Considerations and Laboratory Tests	
10	**SAE J1211**	Handbook for Robustness Validation of Automotive Electrical/Electronic Modules	SAE Website
11	**TPJLR.00.187**	Squeak and Rattle Shaker Tests for Components or Systems	JLR Standards Website
12	**GBT 28046**	National Standard of the People's Republic of China - Road vehicles-environmental conditions and testing for electrical and electronic equipment	
13	**IPC-A-610 Revision F**	Acceptability of Electronic Assemblies	IPC Website
14	**ISO 6270**	Paints and varnishes - Determination of resistance to humidity - Part 2	ISO Website
15	**TPJLR.18.139**	Dewing Test	JLR Standards Website
16	**ISO 17025**	General requirements for the competence of testing and calibration laboratories	ISO Website

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G INTRODUCTION

This document describes the test methods and equipment requirements to be used to verify that electrical and electronic modules (including sensors, actuators and switches) satisfy the component level environmental testing for EC-7128/STJLR.18.233 requirements. The test reference code is indicated in the title of each test procedure.

NOTE: A DUT that does not meet requirements stipulated in the acceptance criteria shall be considered a failure and removed from test at the time of the discovery of the non-compliance. An 8D shall be started and the JLR Component Engineer shall be informed of the non-compliance within 72 hours of the discovery. Failed/non-compliant/suspect DUTs shall not be put back into or allowed to continue the test.

G 1 FACILITIES

The tests shall be conducted at locations having the necessary equipment and facilities with accreditation to ISO 17025.

G 2 APPLICABILITY OF TESTS

Applicability of a test is as specified in EC-7128, STJLR.18.233 or as agreed by the relevant subject matter experts and included in the component PDS. The component/unit/device being tested will be referred to as the Device under Test (DUT).

G 3 ACCEPTANCE REQUIREMENTS

Acceptance of DUT test results shall be based on the evaluations and the performance requirements described in the specific tests, section 0 and applicable Design Verification Methods listed in the Product Design Specification (PDS).

G 4 SAMPLE SIZE

Minimum samples sizes shall be as defined in the test groups in STJLR.18.233 Section H2. The applicable test groups shall be agreed between JLR and the supplier and defined in the component PDS.

NOTE: If reworked parts are to be included in the tests the sample sizes shall be doubled with the additional samples being reworked parts.

Drop tests, K 16.1, K 16.2, require 14 samples and the Chemical Resistance test K 22, may require additional samples depending on the number of chemicals required. Additional samples maybe introduced into the test sequence without pre-conditioning for both Drop and Chemical Resistance tests.

Additional samples may be stipulated by the responsible Component Engineer if there are legal or process quality requirements that dictate their use. HALT (K 28) testing may require a larger sample size if the determination of destruct limits is stipulated.

G 5 SEQUENCE OF TESTS

Tests shall be completed in the order specified in the test groups in STJLR.18.233 Section H2, or as agreed between the supplier and the JLR Component Owner, JLR D&R team and JLR Environmental Technical Specialists. The applicable test groups shall be defined in the component PDS.

G 6 MICROPROCESSOR REQUIREMENTS

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The DUTs must have a production representative microprocessor/memory device. In addition, a vehicle representative software strategy shall be used. The software should be in a normal operating mode (i.e. not in idle or standby mode). Test software is allowable for DV testing but not for Product Validation (PV).

G 7 RETENTION OF TEST SAMPLES

Samples used for DV evaluation, PV evaluation, testing, or held as 'golden sample' during the tests shall be available for JLR inspection/evaluation for one year after the Job 1 date for each vehicle line application.

G 8 ORIENTATION OF DUTs DURING TESTING

Where specified in the individual test methods, DUTs shall be mounted in the test chamber in the same orientation as used in the vehicle. Where multiple packaging orientations are possible, either the worst case or multiple test orientations shall be agreed between the JLR Component Engineer, Supplier and Test House.

G 9 PROVISION OF TEST REPORTS AND SUPPLEMENTARY DATA

Testing shall be accompanied by completed test reports in English. Test reports shall be delivered to the JLR Component Engineer. Photography shall be of sufficient resolution and quality to see the features of interest, see IPC-A-610 Revision F (Section 1.8, 1.10 and 1.11) as a guide to an acceptable standard.

Supplementary data (photographs, tables, text, video, audio etc.) shall be available to Jaguar Land Rover.

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H INSTRUMENTATION

H 1 GENERAL TEST

Setup and Fixtures.

H 1.1 The device under test shall be tested with the subsystem or system configuration, as agreed with the JLR Component Owner.

H 1.2 Grounding. Unless otherwise specified, all inputs and outputs, including power, shall be referenced to a single-point ground.

H 1.3 Temperature. The device under test (and mounting fixture if used) will be completely surrounded by an envelope of air. The temperature of the envelope of air shall be within 2 °C of the test temperature. This temperature is to be measured and verified 2.5 cm from the DUT at non-operating conditions. Pre-test temperature balance data shall be recorded in the test log.

H 1.4 Pre & Post Test Checks. Acceptance tests according to section G 3 and section L shall be completed prior to and immediately after each test .These data along with the component identification code, test identification number and test history shall be recorded.

A test log or record shall be created that includes:

- The input/output signals monitored during and after each test. Key parameters to be monitored are defined in section L. Permissible degradation levels shall be defined by the JLR Component Owner.
- Test equipment identification, including but not limited to identification of thermal/salt/dust/water chambers; vibration test equipment along with temperature balance data recorded before the test commences.
- Recorded data.
- Discrepancies and 8Ds.
- Post-test sign-off by the test and design engineers.
- Temperature monitoring and acceleration. Identify the point or points on the module/board that will be used to monitor temperatures/accelerations for the various tests in section K.

NOTE: A copy of the signed test log shall accompany the D/PVP&R provided to the JLR Component Engineer. This shall be signed either physically or digitally.

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H 2 INTERCONNECTIONS

DUTs shall be connected in an electrical subsystem configuration to represent the vehicle system as much as is practical.

Test fixtures shall include use of production connectors, if available, in such a way that they will be made and not broken, e.g. pigtails, until the completion of testing on the DUTs has reached the Connector Lead/Lock test. This will help identify connector interface problems.

Connecting harnesses shall be long enough to leave the chamber or other such test setup without the requirement for additional connectors or splices. The electrical properties of the test harness shall be agreed with the JLR Component Engineer.

H 3 INPUT AND OUTPUT TERMINATIONS

DUT inputs and outputs shall be attached to the actual loads used in the vehicle configuration (solenoids, motors, coils, relays, switches etc.). Where this is not feasible, devices (or loads) that electrically simulate the device may be used if agreed with the JLR Component Engineer.

H 4 POWER SUPPLY REQUIREMENTS

Supply voltage shall meet the tolerances listed in Section I. Voltage levels shall be specified in the PDS. Section M 1 is provided for information.

H 5 CALIBRATION

Test measurement equipment must have valid calibration certificates and maintenance schedules available for inspection.

H 6 SAFETY

Applicable safety guidelines and procedures must be followed.

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I EQUIPMENT and FACILITIES

Humidity/thermal chamber shall meet the thermal loading, temperature and humidity control requirements listed below.

Vibration equipment shall meet the frequency and acceleration requirements listed below.

Power supply equipment shall meet the control requirements listed below. Power supplies shall be stable throughout the full range of DUT operating currents and voltages.

Equipment maintenance records shall be available for review at the request of JLR.

I 1 TOLERANCES

Unless otherwise stated, tolerances shall be:

- Voltage: V_{min}, V_{max}, V_{pmin} and V_{pmax} ±0.2 V, V_N ±0.5 V
- Frequency: ±2 %, 0.5 Hz below 25 Hz
- Acceleration: ±10 %.
- Time: ±2 %.
- Temperature: ±3 °C

I 2 ACCURACY OF TEST INSTRUMENTATION.

I 2.1 To ensure compliance to specified limits proper allowance for equipment measurement errors (including inaccuracies) shall be taken into account.

I 2.2 The accuracy of measurement equipment shall be better than $^1/_{10}{}^{th}$ of the tolerance of the variable to be measured.

I 2.3 Thermal chamber loading. Mass loading profiles shall not exceed the manufacturers maximum specifications. The estimated loading factors shall be listed in the test log for each piece of equipment used.

I 3 FACILITIES

Unless within a specified test or acceptance process, the DUTs shall be kept under Room Temperature and Pressure (RTP) conditions where air temperature lies in the range (15 to 35) °C, relative humidity (25 to 75) % and air pressure (86 to 106) kPa as per IEC 60068-1-1 Section 5.3 Standard atmospheric conditions for measurement and tests.

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J GENERAL TEST REQUIREMENTS

The applicable test groups in STJLR.18.233 shall be defined and recorded in the PDS. Each of the DUTs shall be subjected to the specific tests contained within the test group(s).

The following test temperatures, test voltages and other parameters shall be defined and cascaded in the PDS.

- V_{min}, V_{max}, V_{pmin}, V_{pmax}, V_N – Section M 1.
- T_{min}, T_{max}, T_{Pmin}, T_{Pmax}, T_{Emin}, T_{Emax}, T_{soak} – Section M 2.
- Water/Fluid Ingress – Section M 3.
- Salt and Chemical Resistance – Sections M 4 and M 5.
- Vibration in tests – Section K 15.
- Mechanical Shock/Drop in test K 16.

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K OPERATION

K 1 **CL-LTE** - LOW TEMPERATURE EXPOSURE

ISO 16750-4:2010 Section 5.1.1.1

IEC 60068-2-1:2007 (Test Ab)

This test is to verify that the component can withstand the effects of extreme low temperatures encountered in vehicles used in cold climates and the cold temperatures that may be experienced in shipping.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-1 must be listed in the test facility's accreditation schedule.

Possible effects - changes in the physical properties of materials resulting in:

- Hardening and embrittlement
- Binding or loosening due to contraction of dissimilar materials
- Cracking/crazing
- Loss of lubrication and lubricant flow
- Static fatigue
- Condensation and freezing of water.

K 1.1 Equipment and Facilities

The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I.

Forced air circulation may be used to maintain homogeneous conditions.

K 1.2 Test Setup And Preparation

Mount the DUTs in the chamber at the same orientation as used in the vehicle.

K 1.3 Procedure

a Power up the chamber and stabilize the DUTs at a temperature of -40 °C (T_{Emin}). The rate of temperature change in the chamber shall not exceed 1 °C/min (averaged over a 5 minute period).

b Expose the DUTs for 24 hours. The DUTs shall be unpowered. Harnesses and monitoring instrumentation may be connected as agreed with the Component Engineer.

c When the test is complete perform the Functional Check (L 2) at -40 °C (T_{Emin}) and Visual Check (L 5) If the DUTs pass proceed to the next test.

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K 2 **CL-HTE** - HIGH TEMPERATURE EXPOSURE

ISO 16750-4:2010 Section 5.1.2.1

IEC 60068-2-2:2007 (Test Bb)

This test is to verify that the component can withstand the effects of high temperatures which occur in the vehicle as a result of high ambient temperatures combined with solar radiation and/or powertrain heat rejection.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-2 must be listed in the test facility's accreditation schedule.

Possible effects - changes in the physical properties of materials resulting in:

- Binding due to expansion of dissimilar materials
- Reduction in viscosity of lubricants. Outward flow of lubricants from seals
- Distortion or deterioration of seals, gaskets etc.
- Permanent setting of resinous materials such as plastics and rubber
- High internal pressures created in sealed parts
- Cracking or crazing of organic materials.

K 2.1 Equipment and Facilities

- A climatic chamber capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.
- Equipment required to monitor and verify correct DUT function.

K 2.2 Test Setup and Preparation

Mount the DUTs in the chamber at the same orientation as used in the vehicle. Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

K 2.3 Procedure

a Power up chamber and stabilize at the specified temperature T_{Emax}. Exposure the DUTs for 24 hours.

b The rate of temperature change in the chamber shall not exceed 1 °C/min averaged over a 5 minute period.

c When the test is complete perform the Functional Check (L 2) and Visual Check (L 5) at T_{Emax}. If the DUTs pass proceed to the next test.

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K 3 **CL-LTO** - LOW TEMPERATURE OPERATION

ISO 16750-4:2010 Section 5.1.1.2

IEC 60068-2-1:2007 (Tests Ad, Ae)

This test is to verify that the component operates properly at low temperatures encountered during vehicle cold weather operation.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-1 must be listed in the test facility's accreditation schedule.

Possible effects - changes in physical properties of materials resulting in:

- Hardening and embrittlement
- Binding or loosening due to contraction of dissimilar materials
- Cracking/crazing
- Loss of lubrication and lubricant flow
- Static fatigue
- Condensation and freezing of water
- Changes in electronic components and performance of electronic circuits
- Changes in performance of transformers and electro-mechanical components.

K 3.1 Equipment and Facilities

A climatic chamber capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I. The velocity of air circulation shall be determined from the requirements of IEC 60068-2-1.

Equipment required to monitor and verify correct DUT function.

K 3.2 Test Setup and Preparation

Mount the DUTs in the chamber at the same orientation as used in the vehicle.

Connect the DUTs, harnesses and any monitoring instrumentation required and verify that the DUTs are functioning properly.

Power shall be cycled to provide the opportunity for inrush failures; the duty cycle shall be supplied by the JLR Component Engineer.

K 3.3 Procedure

a Power up chamber and stabilize at temperature T_{min}. The rate of temperature change in the chamber shall not exceed 1 °C/min (averaged over a 5 minute period).

b Perform Monitored Operation on the DUTs as specified in section L 3 during the 24 hour exposure period. Unless otherwise agreed between the Supplier and the JLR Component Engineer the operation shall include periods of Normal Load Stress and Maximum Load Stress.

NOTE: Power shall be cycled to maintain board and component temperature as close to T_{min} as possible and to provide the

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opportunity for inrush failures. A duty cycle of 1 minute on - 4 minutes off is a possible starting point.

c When the test is complete perform the Functional Check (L 2) and Visual Check (L 5) at T_{min}. If the DUTs pass proceed to the next test.

K 4 **CL-HTO** - HIGH TEMPERATURE OPERATION

ISO 16750-4:2010 Section 5.1.2.2

IEC 60068-2-2:2007 (Test Bd, Be)

NOTE: Where the High Temperature Endurance (HTE) test is to be run this test may be waived. Written agreement from the JLR Component Engineer shall be sought.

This test is to verify that the component operates properly at high temperatures which occur in the vehicle as a result of high ambient temperatures combined with solar radiation and/or vehicle self-heating.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-2 must be listed in the test facility's accreditation schedule.

Possible effects - changes in physical properties and/or dimensions of materials resulting in:

- Binding due to expansion of dissimilar materials.
- Reduction in viscosity of lubricants.
- Outward flow of lubricants from seals.
- Distortion or deterioration of seals, gaskets etc.
- Permanent setting of resinous materials such as plastics and rubber.
- High internal pressures created in sealed parts.
- Changes in the stability and/or performance characteristics of electronic circuits and components.
- Overheating of transformers, electro-mechanical components and power devices.
- Altering the operation/release margins of relays and magnetic or thermally activated devices.

K 4.1 Equipment and Facilities

A climatic chamber capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I. The velocity of air circulation shall be determined from the requirements of IEC 60068-2-2. Additional requirements regarding the airflow may be prescribed in the PDS.

Equipment required to monitor and verify correct DUT function.

K 4.2 Test Setup and Preparation

Mount the DUTs in the chamber at the same orientation as used in the vehicle.

Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

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K 4.3 Procedure

a Power up chamber and stabilize at the specified temperature T_{max}/T_{pmax}. Perform Monitored Operation on the DUTs as specified in section L 3 during the 96 hour exposure period. Unless agreed between the Supplier and the JLR Component Engineer, Normal Load Stress conditions shall be used.

b The rate of temperature change in the chamber shall not exceed 1 °C/min averaged over a 5 minute period.

c When the test is complete perform the Functional Check (L 2) and Visual Check (L 5) at T_{max}/T_{pmax}. If the DUTs pass proceed to the next test.

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K 5 **CL-TST** - TEMPERATURE STEP TEST

ISO 16750-4:2010 Section 5.2

This test is to verify that the component is able to operate at all temperatures within its temperature specification.

Possible effect - intermittent or erroneous operation.

K 5.1 Equipment and Facilities

The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I.

Equipment required to monitor and verify correct DUT function.

K 5.2 Test Setup and Preperation

Mount the DUTs in the chamber at the same orientation as used in the vehicle

Connect the monitoring instrumentation and verify that the DUTs are functioning properly

K 5.3 Procedure

NOTE: The DUT shall be in sleep mode or if there is no sleep mode powered OFF during the temperature transition and stabilisation phases.

a Power up the chamber and stabilise at 20 °C. Perform a Functional Check as defined in Section L 2 at Maximum, Minimum and Nominal voltage.

b Lower the chamber temperature by 5 °C, and allow the temperature of the DUT to stabilise.

c Perform a Functional Check as L 2 at Maximum, Minimum and Nominal voltage.

d Repeat steps (b) and (c) until -40 °C is reached.

e Raise the chamber temperature by 5 °C and allow the temperature of the DUT to stabilise.

f Perform a Functional Check as L 2 at Maximum, Minimum and Nominal voltage.

g Repeat steps (e) and (f) until T_{max} is reached.

h Lower the chamber temperature by 5 °C, and allow the DUT temperature to stabilise.

i Perform a Functional Check as L 2 at Maximum, Minimum and Nominal voltage.

j Repeat steps (h) and (i) until 20 °C is reached.

k When the test is complete perform a Visual Check (L 5).

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K 6 **CL-PTC** - POWERED THERMAL CYCLE

ISO 16750-4:2010 Section 5.3.1

IEC 60068-2-14:2009 (Test Nb)

This test is to verify that the component is able to operate in the changing temperature conditions encountered in the vehicle environment.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-14 test Nb must be listed in the test facility's accreditation schedule.

Possible effects:

- Breakage or cracking due to stresses created by expansion and contraction of dissimilar materials.
- Intermittent or erroneous operation.

K 6.1 Equipment and Facilities.

The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I.

Equipment required to monitor and verify correct DUT function.

K 6.2 Test Setup and Preparation.

Mount the DUTs in the chamber at the same orientation as used in the vehicle.

Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

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K 6.3 Procedure.

a For units that have a continuous and peak maximum operational temperature the alternative profile in Figure 2 shall be used.

b Power up chamber and run the profile in Figure 1/Table 2 or Figure 2/Table 3. Expose the DUTs for 30 cycles. Additional cycles may be prescribed in the PDS.

Power shall be cycled to provide the opportunity for inrush failures; the duty cycle shall be supplied by the JLR Component Engineer.

NOTE: The humidity of the chamber shall be regulated to ensure that no condensation forms on the DUTs.

c Monitored operation as described in L 3 shall be performed in the operational sections. Component operation shall be cycled between Normal Load Stress and Maximum Load Stress.

When the test is complete perform the Functional and the Visual Check as specified in L 2 and L 5 at T_N. If the DUTs pass proceed to the next test.

Figure 1 - Powered Thermal Cycle Profile 1

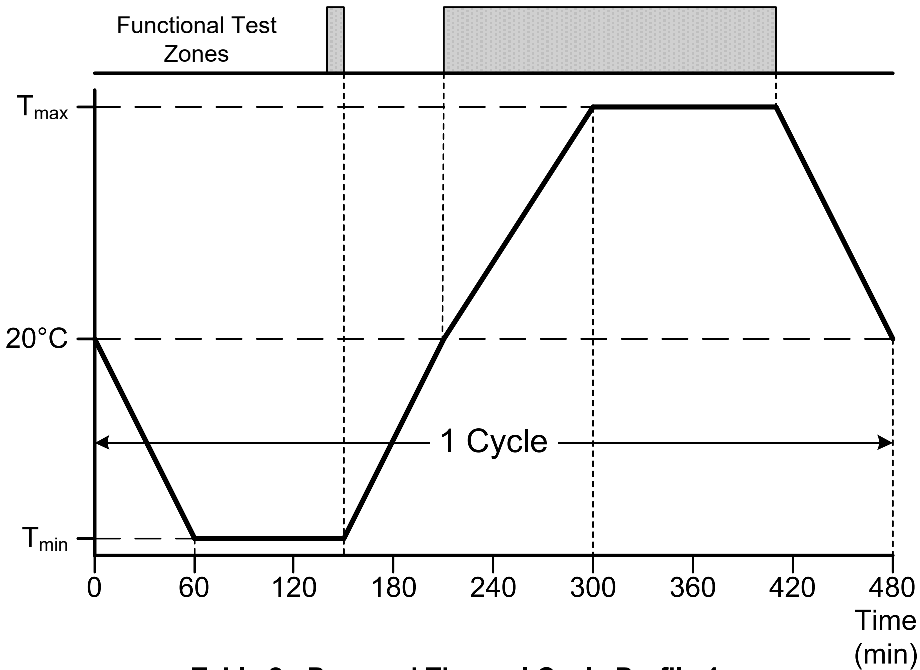

Table 2 - Powered Thermal Cycle Profile 1

Time (Min)	Temperature (°C)
0	20
60	T_{min}
150	T_{min}
210	20
300	T_{max}
410	T_{max}
480	20

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Figure 2 - Powered Thermal Cycle Profile 2

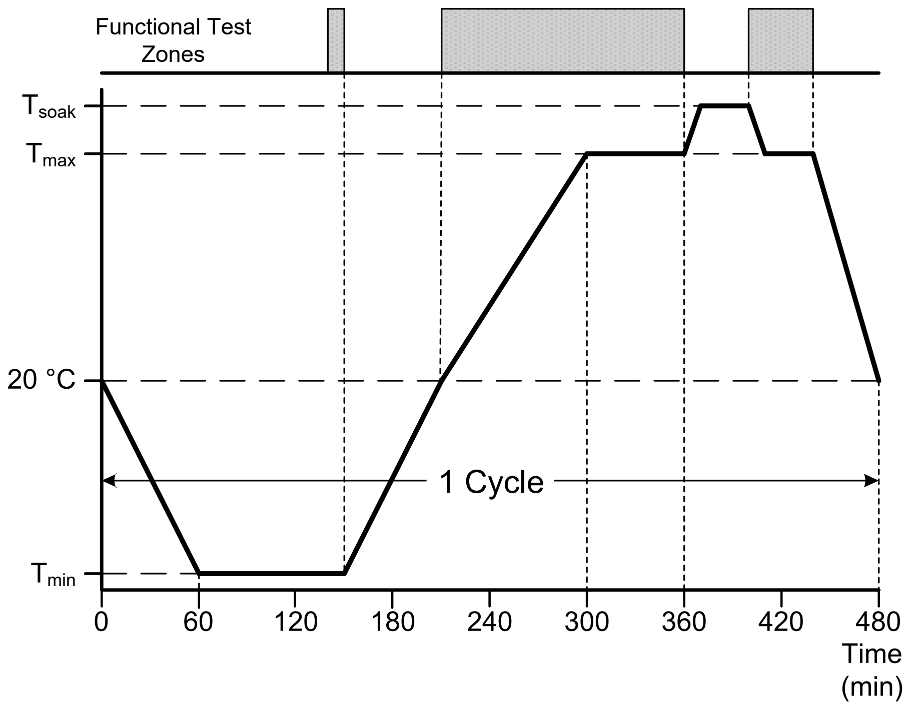

Table 3 - Powered Thermal Cycle Profile 2

Time (Min)	Temperature (°C)
0	20
60	T_{min}
150	T_{min}
210	20
300	T_{max}
360	T_{max}
370	T_{soak}
400	T_{soak}
410	T_{max}
480	20

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K 7 **CL-TSA** - THERMAL SHOCK IN AIR

ISO 16750-4 Section 5.3.2

IEC 60068-2-14:2009 Test Na

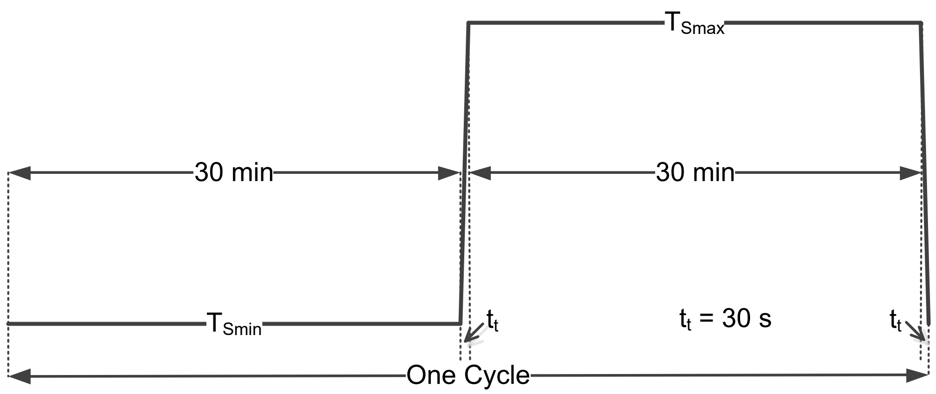
This test is to verify that the component can withstand the effects of rapid changes in ambient air temperature which occur in the vehicle environment and during shipping and handling.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-14 test Na must be listed in the test facility's accreditation schedule.

Possible effects:

- Binding or loosening of moving parts
- Deformation, cracking or fracture of materials
- Leaking of sealed compartments
- Changes in physical alignment of related parts and optical equipment.

Figure 3 - Thermal Shock in Air (Profile)



K 7.1 Equipment and Facilities

The chambers shall be capable of maintaining the specified temperature(s) in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.

The thermal shock system shall consist of two thermal chambers with an automated mechanical system to move the test samples from one chamber to the other. Alternatively, a chamber capable of transitioning from T_{max} to T_{min} in less than 30 seconds and from T_{min} to T_{max} in less than 30 seconds may be used.

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K 7.2 Test Setup and Preparation

NOTE: Where a component is packaged close to a heat source complete 300 cycles, otherwise 100 cycles shall be completed.

NOTE: If logistics require the test to be interrupted then the DUT shall be maintained at either T_{min} or T_{max} until the test can be resumed.

Mount the DUTs at the same orientation as used in the vehicle. It is highly recommended that the DUTs be rack-mounted, not just placed on the floor of the chamber.

The DUTs shall be unpowered. Harnesses and monitoring instrumentation may be connected if agreed with the JLR Component Engineer.

K 7.3 Procedure

a Bring both chambers to their respective set temperatures T_{Emin} and T_{Emax} as indicated in Figure 3.
b Move the DUT's into the cold chamber within 30 seconds and start the cold soak cycle. Soak the DUTs for 30 minutes.
c Move the DUTs into the hot chamber within 30 seconds and begin the hot soak cycle. Soak the DUTs for 30 minutes.
d Repeat steps (b) and (c) for the required number of cycles.
e Perform Functional Check (L 2) and Visual Check (L 5).
f If the DUTs pass proceed to the next test.

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K 8 **CL-DEW** - DEWING TEST

ISO 16750-4:2010 Section 5.6.2.4 Test 3

This test is to reveal design deficiencies caused by condensation on electrical and electronic circuits within the module.

Possible effects are electrical failures as a result of:

- Increased leakage currents
- Electrochemical migration (dendritic growth) and electrical short circuits
- Corrosion.

K 8.1 Equipment and Facilities

K 8.1.1 A vapour-tight climatic test chamber capable with the following attibutes.

a It shall be capable of generating a condensation atmosphere with controlled humidity and air temperature as described in BS EN ISO 6270-2:2005 Section 5.

b It shall be equipped with corrosion-resistant inner walls that are kept free of instances of solids or liquid constituents that could contaminate the condensate during testing and affect the DUTs.

c It shall be equipped with a temperature-controlled water bath or water filled floor trough which shall be kept filled with distilled water (maximum conductivity 5 µS/cm) to a depth of no less than 10 mm at all times during operation.

d Condensed water shall be continuously drained from the chamber and shall not be re-used.

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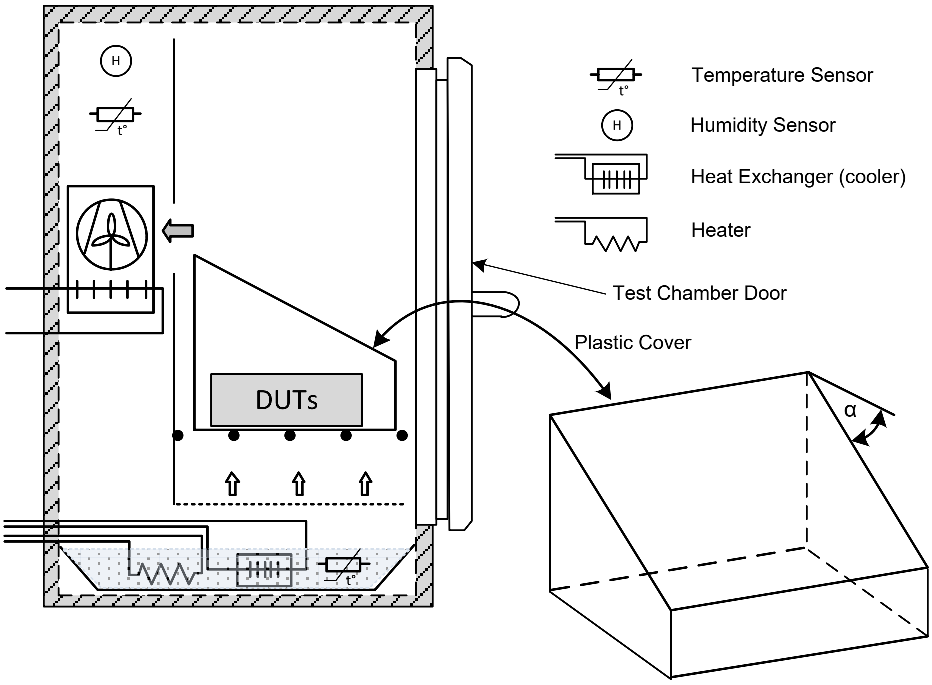
K 8.1.2 A plastic cover (see Figure 4) to eliminate the effects of chamber forced air circulation as defined below:

a It shall have 4 sides and a sloping roof with an inclination ≥12 ° to prevent condensation dripping on DUTs.

b It shall be open at the bottom and positioned on a floor grid (≥200 mm above the surface of the water) with its roof sloping towards the chamber door.

c The cover shall have its dimensions adapted to the size of the test area of the climatic chamber. It shall be mounted no closer than 8 cm or 10% of the dimension (width, height, depth), whichever is greater, to any wall.

Figure 4 - Dewing Chamber



The inclination of the plastic cover roof, α must be ≥ 12°

K 8.1.3 Support devices to maintain installation orientation of the DUTs as below:

- Parts of the supporting devices that come into contact with the DUTs shall be constructed of material that will not cause electrolytic corrosion.
- The supporting devices shall have low thermal conduction (to ensure thermal isolation of the DUTs).

K 8.1.4 Equipment to measure and log:

a Chamber temperature and chamber humidity

b Water bath temperature

c DUT surface temperature

d Correct DUT function, current consumption and voltage.

K 8.2 Test Setup and Preparation

The 6 DUTs are to be tested with as much of the housing removed as possible, without compromising normal operation, in order to maximise the exposure of the PCBA(s) to the condensation.

a Connect the DUTs, harnesses and any monitoring instrumentation required and verify that the DUTs are functioning properly.

b The wiring harnesses must protrude beyond the test chamber without further connections.

c The connectors shall remain connected at all times during testing.

d Two of the DUTs shall be installed horizontally (with primary side up), two inverted (with primary side down) and the final two DUTs shall be oriented as installed in the vehicle.

e The DUTs shall be positioned ≥ 100 mm from the walls, ≥ 200 mm above the surface of the water and shall be spaced ≥ 20 mm from adjoining DUTs.

K 8.3 Test Procedure

NOTE: If logistics require the test to be interrupted then the test shall be paused at the start of a cycle under the initial test conditions (20 °C ± 3 °C, 100% RH (+0, -5) %RH).

Pre-conditioning phase

Not shown in Figure 5.

a Introduce the DUTs into the chamber set at the standard atmospheric conditions for measurement and tests.

b Set the temperature and humidity in the chamber to the initial test conditions. (20 °C ±3 °C, 100% RH (+0, -5) %RH)

c The initial test conditions are maintained for a total duration of 45 minutes (to ensure DUT temperature stability prior to the start of the first test cycle).

Test Cycle

After the pre-conditioning phase, when not being functionally tested, the DUTs shall be placed in Sleep mode (at 12 V ± 0.2 V) or if there is no Sleep mode available shall be switched off.

a The supply voltage and current consumption of the DUTs shall be measured and logged throughout the test.

b Complete IEC 60068-2-30 Test Db using the variant in Figure 5. The condensation phase may be shortened if required to maintain a difference of 1 °C to 2 °C between the surface of the DUT and the chamber to ensure that condensation forms on its surface.

c During the condensation phase the ambient temperature of the chamber shall be controlled by heating the water in the bath/floor trough. The temperature difference between the water bath and the ambient temperature of the chamber must always be <15 °C. The test chamber wall temperature must be higher than the DUT temperature.

d A short functional test at 13.5 V ±0.2 V (between 15 seconds and 2 minutes duration) shall be performed on the DUTs every 10 °C rise in temperature (at 30 °C, 40 °C, 50 °C, 60 °C) as shown in Figure 5.

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NOTE: The functional test shall be designed to minimize self-heating.

- e A long functional test at 13.5 V ± 0.2 V shall start when the DUTs reach 70 °C and continue for 15 minutes after reaching 80 °C.
- f After the condensation phase switch to chamber temperature control and maintain the chamber temperature at 80 °C for 45 minutes.
- g During the drying phase switch off the water bath heater and use the chamber temperature control to return the temperature to initial conditions (20 °C ±3 °C) over a 90 minute period.
- h Repeat from c for 5 cycles.
- i When the test is complete perform Functional Check (L 2) starting with the parameters most sensitive to moisture followed by Visual Check (L 5). The DUT function shall be Class A according to ISO 16750-1. Where the function provided by the DUT is convenience only, this may be reduced to Class C (when agreed between the JLR component owner and the supplier).

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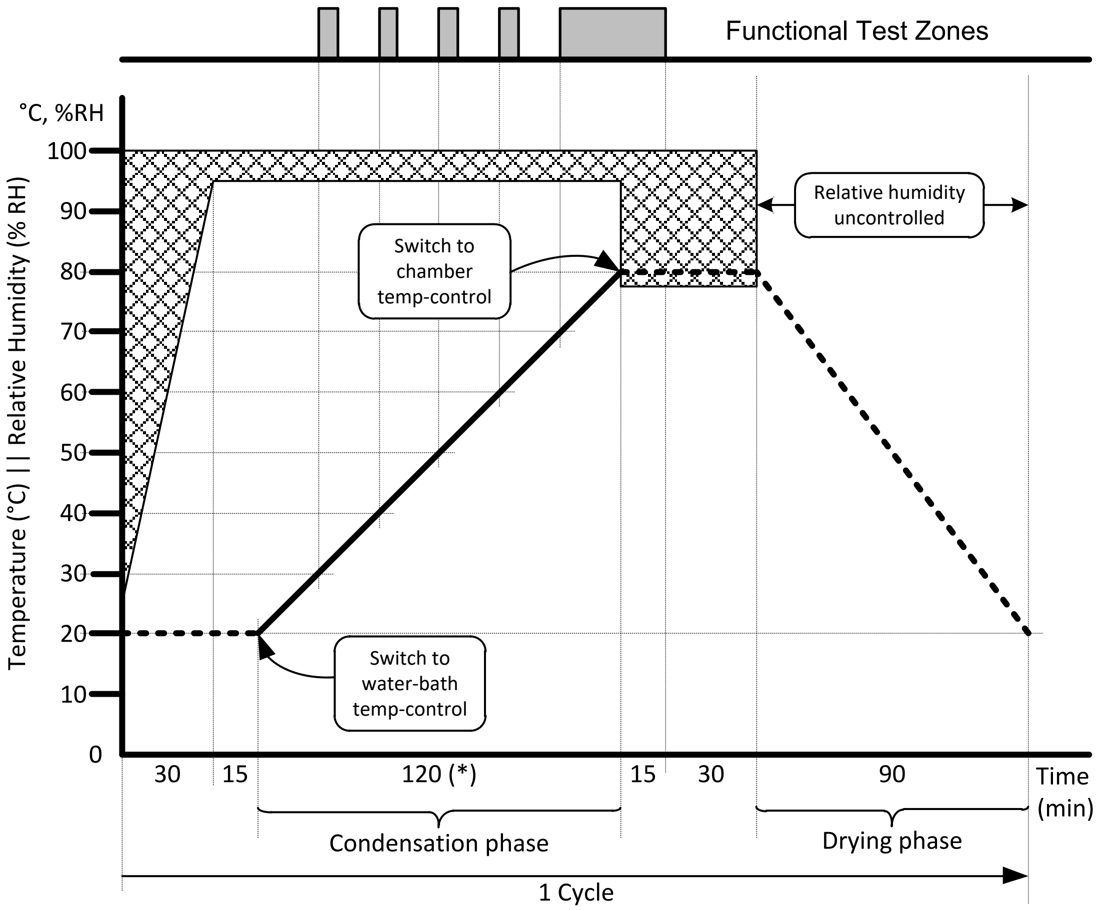
Figure 5 - Dewing Test Profile



(*) Condensation phase duration (nominally 120 minutes) will vary to meet the requirement for DUT temperatures to be 1 °C to 2 °C below the chamber temperature.

Key/Legend

Functional Test Zone	Test chamber humidity
Test chamber temperature ±3 °C (chamber temperature controlled)	Water-bath temperature ±1 °C (water-bath heater controlled)

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K 9 **CL-THC** - TEMPERATURE/HUMIDITY CYCLE

ISO 16750-4 Section 5.6.2.3

IEC 60068-2-38 Test Z/AD

This test is to verify there are no deficiencies in the design of the electrical/electronic module caused by "breathing", including the effect of trapped water freezing leading to cracks and fissures and the effects of frosting.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-38 must be listed in the test facility's accreditation schedule.

Possible effects include physical and chemical deterioration of materials leading to:

- Swelling of hygroscopic materials
- Degradation in electrical and thermal properties in insulating materials
- Electrical short circuits
- Binding of moving parts
- Oxidation and/or galvanic corrosion of metals
- Degradation of electrical components/circuits
- Degradation of image transmission through optical elements.

K 9.1 Equipment and Facilities

The chamber shall be capable of maintaining the specified temperature and humidity in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.

Equipment required to monitor and verify correct DUT function.

Additional equipment required to measure and log:

- Chamber temperature and humidity.
- DUT voltage and current consumption.

K 9.2 Test Setup and Preparation

Connect the DUTs, harnesses and any monitoring instrumentation required and verify that the DUTs are functioning properly.

DUTs shall complete a Visual Check (L 5) and Functional Check (L 2) prior to testing

Mount the DUTs in the chamber at the same orientation as used in the vehicle.

K 9.3 Procedure

a Complete the pre-conditioning cycle according to IEC 60068-2-38 Test Z/AD

b The DUTs shall then be subjected to the temperature/humidity cycle shown in Figure 6.

c During this cycle, the DUTs shall be in Sleep mode if available (at 12 V ±0.2 V) or switched off.

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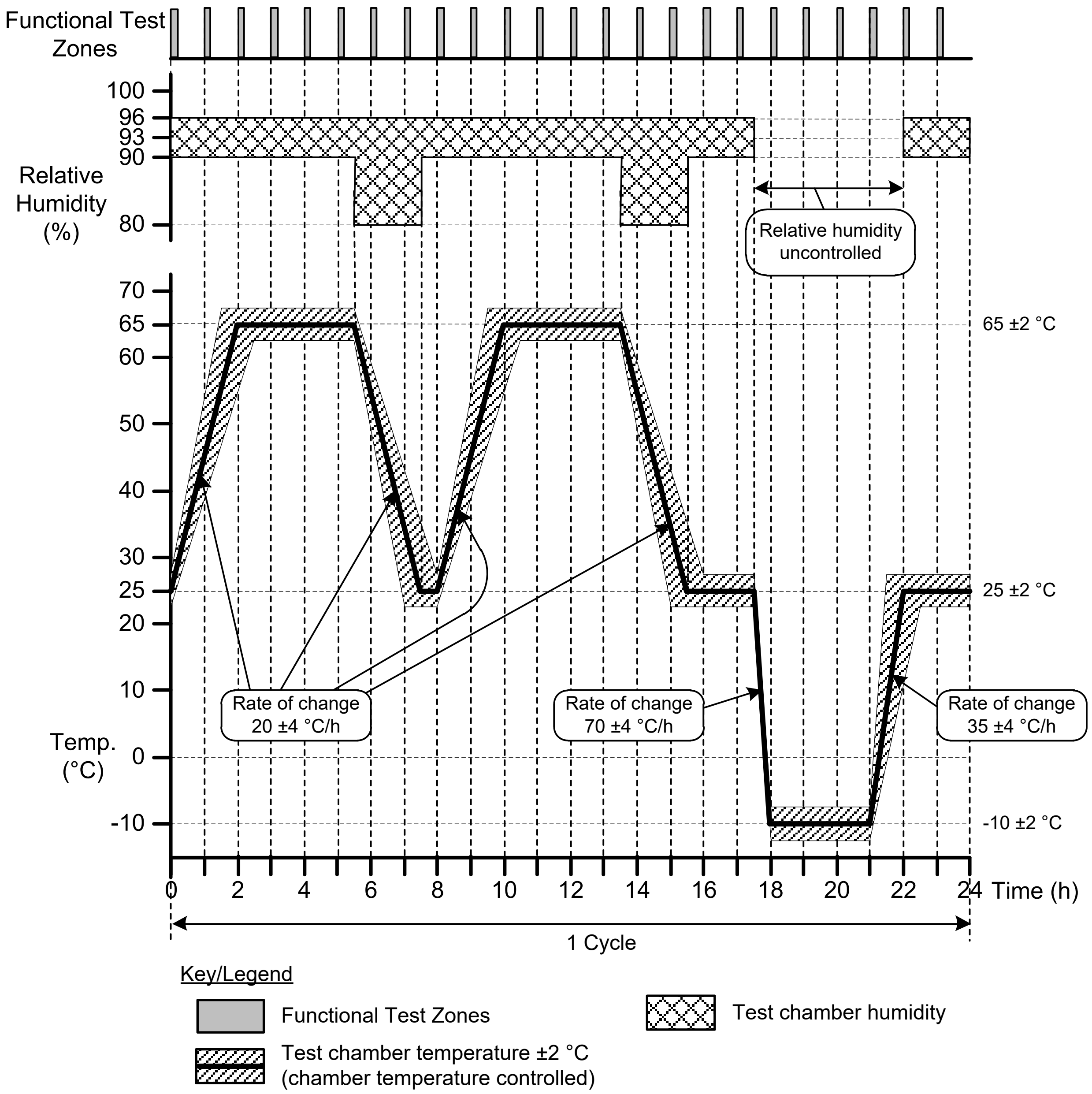
d The DUT shall be powered on and a short Functional Check (L 2) performed during that last hour of each cycle.
e The temperature/humidity cycle shall be repeated for 10 cycles.
f There shall be a final 24 h drying period as specified in IEC 60068-2-38 Test Z/AD
g Acceptance criteria: When the test is complete return the temperature to T_N and perform the Functional Check (L 2) and Visual Check (L 5)

Figure 6 - Temperature/Humidity Cycle Test Profile



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K 10 **IN-W** - WATER/FLUID INGRESS TESTS (IPX1 – IPX9K)

NOTE: The applicable water/fluid ingress ratings shall be defined in the PDS

Moving parts (e.g. motors, switches) shall be operated during testing, unless otherwise stated in the PDS.

Where the IP rating is dependent on orientation, this shall be documented in the PDS and in the Test Report. The test report shall include photography of the test orientation.

ISO 20653

Water/fluid ingress tests are to verify the component can withstand the effects of the moisture exposure which can occur in its packaging environment.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to ISO 20653 must be listed in the test facility's accreditation schedule.

Possible effects:

- Electrochemical migration (Dendrites) causing malfunction/resulting in an unsafe condition
- Corrosion due to direct exposure/high humidity levels caused by the water
- Damage to seals
- Removal/deterioration of required labels
- Loss of lubricants between moving parts.
- Degradation in Surface Insulation Resistance.

K 10.1 Equipment and Facilities

In addition to that required in ISO 20653.

- A means of verifying the water flow rate.
- A means of verifying the temperature of the water.
- A means of detecting water ingress such as a tracer added to the water or water detecting test strips unless this is not technically feasible.

K 10.2 Test Setup and Preparation

In addition to that required in ISO 20653.

- Where it is required to meet the IP rating connectors shall be mated.
- Where the DUT has exterior moving parts, these parts shall be in motion unless specifically prohibited by agreement between JLR and the Supplier and stated in the PDS.

K 10.3 Procedure

NOTE: The DUTs may be operational or non-operational as specified by the JLR Component Engineer.

a Perform the test according to ISO 20653

b When the test is complete perform the Functional Check (L 2) and Visual Check (L 5).

c Repeat the test for each mounting orientation.

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d Additional pass criteria: No water is allowed inside the DUT, unless stated in the PDS. If the DUTs pass, continue to the next test.

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K 11 **IN-TSI** - ICE WATER IMMERSION

ISO 16750-4 Section 5.4.3

This test is to verify that the component can withstand the effects of complete immersion including sudden immersion into cold water when operating at a high temperature.Possible effects:

- Cracking or separation of bonded materials such as case and potting
- Damage to seals
- Electrochemical migration (Dendrites) causing malfunction/resulting in an unsafe condition
- Corrosion due to direct exposure/high humidity levels caused by the water
- Removal/deterioration of required labels
- Loss of lubricants between moving parts.

This test shall be completed on components mounted below the wade line on the exterior of the vehicle.

NOTE: For PV testing, if the case tool has multiple cavities, a representative number of parts must be tested from each. Two or three from each would be a recommendation; no less than one each is acceptable. If additional parts need to be added at this test step it is acceptable to do so without pre-conditioning.

K 11.1 Equipment and Facilities

- The water used shall have the addition of 5% chemically pure NaCl (Sodium Chloride) by mass. The water temperature shall be 0 °C +5/-0.
- A means of measuring the water temperature.
- A tank with sufficient depth to completely submerge the DUT with its lowest point being at least 150 mm below the surface of the water.
- The climatic chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I.

K 11.2 Test Setup and Preparation

Mount the DUT in the holding fixture.

A tracer may be added to the water bath to improve the detection of water inside the part.

K 11.3 Procedure

a Place the DUT in the climatic chamber and stabilise the temperature at T_{max} with the DUT operating at Maximum Load. The stabilisation time shall be a minimum of 1 hour or the time taken for the surface temperature of the DUT to achieve the chamber set temperature ±5 °C plus 15 minutes.

b Remove the DUT from the chamber and immerse it in the water bath, within 30 seconds, with its highest point being at least 15 mm below the surface. Keep the DUT immersed for 5 mins. The DUT may be unpowered, powered, or powered and operating at maximum load, as required by the PDS during submersion.

c Repeat steps (a) and (b) 20 times.

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d When the test is complete perform the Functional Check (L 2), and Visual Check (L 5) (These checks may be waived if this test is immediately followed by IP test K 10).

e Additional pass criteria: No water is allowed inside the DUT unless stated in the PDS. If the DUTs pass, continue to the next test.

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K 12 **IN-TSS** - ICE WATER SPLASH

ISO 16750-4 Section 5.4.2

This test is to verify that the component can withstand the effects of being splashed (including with cold water) when operating at high temperature.

Possible effects:

- Cracking or separation of bonded materials such as case and potting
- Damage to seals
- Electrochemical migration (Dendrites) causing malfunction/resulting in an unsafe condition
- Corrosion due to direct exposure/high humidity levels caused by the water
- Removal/deterioration of required labels
- Loss of lubricants between moving parts.

This test shall be completed on components mounted on the exterior of the car but may be waived if Ice Water Immersion Shock (K 11) is completed.

NOTE: For PV testing, if the case tool has multiple cavities, a representative number of parts must be tested from each. Two or three from each would be a recommendation; no less than one each is acceptable. If additional parts need to be added at this test step it is acceptable to do so without pre-conditioning.

K 12.1 Equipment and Facilities

- The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I.
- Equipment required to keep the DUTs powered throughout the test.
- Equipment required to splash the DUTs, such as a water reservoir, pump and nozzle sufficient to splash the entire DUT. A tracer may be added to the water to aid in the identification of fluid ingress.

K 12.2 Test Setup and Preparation

a Stabilize the temperature of the chamber at T_{max}.

b Connect any equipment to keep the DUTs powered during the test.

K 12.3 Procedure

a Mount the DUT in the chamber at the same orientation as used in the vehicle and stabilise temperature at T_{max} with the DUT operating at Maximum Load.

b Remove the DUT from the chamber and, keeping in the same orientation as in the vehicle, within 20 seconds of removal from the chamber splash the DUT with 3 - 4 litres of water over a 3 second period. The water shall be applied over the full width of the DUT, the temperature of the water shall be 0 °C +4/-0. If the DUT can only be splashed from a specific orientation in its packaged location in the vehicle, this orientation shall be used and documented in the PDS and test report.

c Repeat the test 100 times.

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d When the test is complete perform the Functional Check L 2 and Visual Check L 5. (These checks may be waived if this test is immediately followed by IP test K 10).

e Additional pass criteria: Unless stated in the PDS no water is allowed inside the DUT. If the DUTs pass continue to the next test.

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K 13 **IN-D** - DUST INGRESS

ISO 20653:2013 Section 8.3 (IP5KX, IP6KX)

This test is to verify that the component can withstand the effects of the accumulation of fine dust particles resulting from operation of the vehicle in dusty driving conditions.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IS0 20653 must be listed in the test facility's accreditation schedule.

Possible effects:

- Abrasion/erosion of surfaces
- Penetration of seals
- Degradation of electrical circuits caused by reduced heat dissipation capacity due to clogging, coverage, etc.
- Clogging of openings and filters
- Physical interference of moving parts
- Disruption of optics

K 13.1 Equipment and Facilities

- A chamber capable of meeting the requirements of ISO 20653:2013 Section 8.3.
- Test dust A2 according to ISO 12103-1 (Arizona dust) shall be used unless otherwise stated in the PDS.

K 13.2 Test Setup and Preparation

a Prior to the test, DUTs shall be stabilised to the temperature and humidity test conditions for at least 24 hours.

b The orientation of DUTs shall be agreed between the JLR Component Engineer and the supplier based upon vehicle package location, and the primary direction of dust exposure.

K 13.3 Procedure

a Complete the test according to ISO 20653:2013 using the vertical style (settling) dust chamber.

Operating Mode: DUT shall be operating according to Monitored Operation (L 3). The component operation shall be cycled between Normal Load Stress and Maximum Load Stress.

Number of Cycles: 20

Where agreed between the JLR Component Engineer and the Supplier a horizontal style (blowing) dust chamber may be used.

b When the test is complete perform the Functional Check (L 2) and Visual Check (L 5).

c Additional Pass Criteria: For IP6KX no dust is allowed inside the DUT.

d If the DUTs pass continue on to the next test.

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K 14 **IN-BD** - DUST BLOWING TEST

IEC 60068-2-68 Test Lc

This test is to verify that the component can withstand the effects of the accumulation of fine dust particles resulting from operation of the vehicle in dusty driving conditions.

Possible effects:

- Abrasion/erosion of surfaces
- Penetration of seals
- Degradation of electrical circuits caused by reduced heat dissipation capacity due to clogging, coverage, etc.
- Clogging of openings and filters
- Physical interference of moving parts.

Equipment and Facilities

- Dust chamber capable of meeting the concentration and flow requirements stated below.
- The chamber shall be large enough so that there is a minimum space of 100 mm between the surfaces of the DUTs and from the chamber surfaces.

K 14.1 Test Setup and Preparation

a Mount the DUTs in the test chamber, in the appropriate orientation so the dust flow strikes the module in a representative manner.

b This test shall be done using Monitored Operation as specified in L 3.

c Unless stated in the PDS, the dust used shall be according to ISO 12103-1 A2 (Arizona Dust).

K 14.2 Procedure

a Complete the test according to IEC 60068-2-68 Test Lc with the following parameters:

- Temperature: 25 °C to 35 °C.
- Dust Concentration: 0.1 g/m³ to 0.3 g/m³.
- Dust Velocity: 2.5 m/s ±0.5 m/s.
- Relative Humidity: 0-30 % RH

b Perform the test for 1 hour. Wipe or shake off excess dust and perform the Functional Check (L 2) and Visual Check (L 5).

c Repeat b) 3 times.

NOTE: Non-compliance at the 1 hour inspection is considered a failure. A non-compliance at the 2 or 3 hour inspection indicates a weakness which requires the proposed fix (as a result of the 8D) to be reviewed with the JLR Component Engineer (for direction).

d If the DUTs pass continue on to the next test.

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K 15 **ME-V** - VIBRATION

ISO 16750-3

IEC 60068-2-6 (Test Fc)

IEC 60068-2-64

This test is to verify that the component or component/bracket assembly is capable of withstanding the effects of the vehicle vibration environment.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-64 (for Random or Sine on Random) and IEC 60068-2-6 (for Sine or Sine on Random) must be listed in the test facility's accreditation schedule.

Possible effects:

- Loss of/intermittent electrical connection
- Wire chafing
- Touching/shorting of electrical parts
- Loss of optical alignment
- Excessive electrical noise
- Temporary loss of function due to acceleration forces on components (e.g. due to mass of a diaphragm-spring mechanism)
- False operation of electro-mechanical components (relays etc.)
- Loosening of particles/parts which have become lodged in circuits or mechanisms
- Loosening of fasteners
- Seal deformation
- Component fatigue
- Cracking/rupture
- Excitation of tuned mass harmonic.

NOTE: For PV testing, if the case tool has multiple cavities, a representative number of parts must be tested from each. Two or three from each would be a recommendation; no less than one each is acceptable. If additional parts need to be added at this test step it is acceptable to do so without pre-conditioning.

K 15.1 Equipment and Facilities.

- Vibration-inducing machine and necessary holding fixtures. The machine must be capable of maintaining the acceleration and frequency within the requirements specified in Figure 8, Figure 9, Figure 10, Figure 11, Figure 12 and Figure 13 and tolerances in section I.
- Equipment required to monitor and verify correct DUT function.

K 15.2 Test Setup and Preparation.

- Vibration-inducing machine and necessary holding fixtures shall be characterised prior to installation of the DUTs.
- Install the DUTs on the vibration table / holding fixture. DUTs shall be installed without using vehicle mounts unless specifically agreed with the JLR Component Engineer.

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- Connect the monitoring instrumentation and verify that the DUTs are functioning properly.
- Where accelerometers or laser vibrometery are requested by the component engineer: Measure the frequency response of each DUT in the X, Y and Z axis. Add print out to the test log and include in the test report, highlighting any resonance.

NOTE: Further testing may be requested by the component area based upon the measured data, for example resonance dwells. This testing shall be agreed between the JLR Component Engineer and the Supplier.

K 15.3 Procedure.

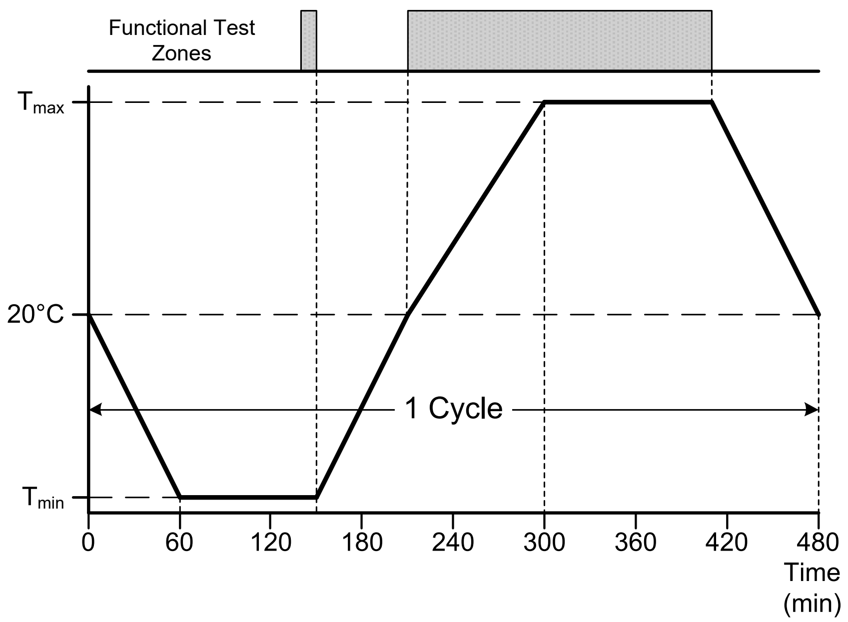
a Complete the applicable profiles below based upon the component packaging location. Test temperatures and operation shall be as Figure 7 unless otherwise agreed with the JLR component Engineer.

b A sweep rate of 0.5 octaves a minute shall be used for sinusoidal tests

c When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue to the next test.

NOTE: Special attention shall be paid to the validation of large / bulky circuit elements such as bulk capacitors or inductors.

Figure 7 - Powered Vibration Temperature Profile

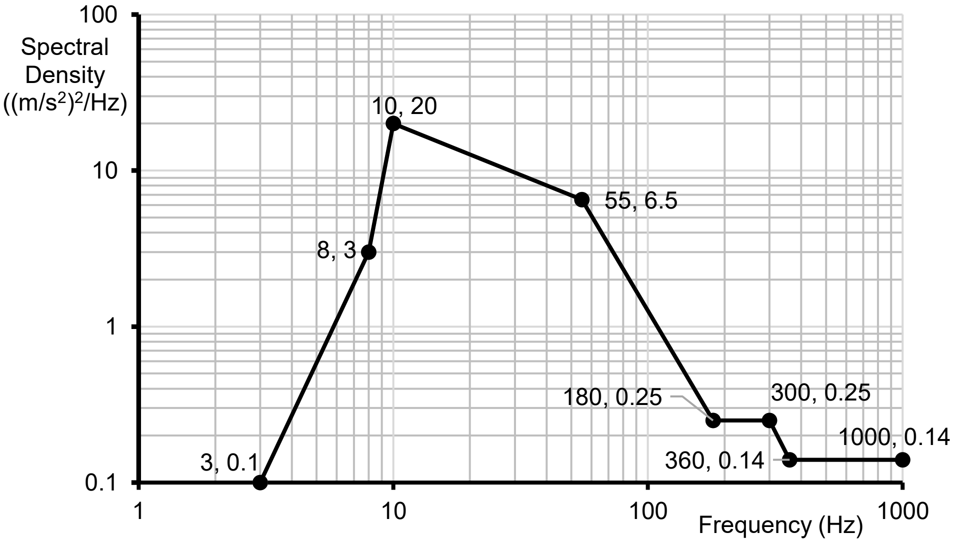


K 15.3.1 Body Mounted (Sprung mass)

Random Wide-Band Random Vibration excitation according to IEC 60068-2-64 using the profile in Figure 8.

Test duration is 8 hours for each axis with RMS acceleration value of 30.8 m/s².

Figure 8 - Vibration Profile - Body Mounted (sprung mass)

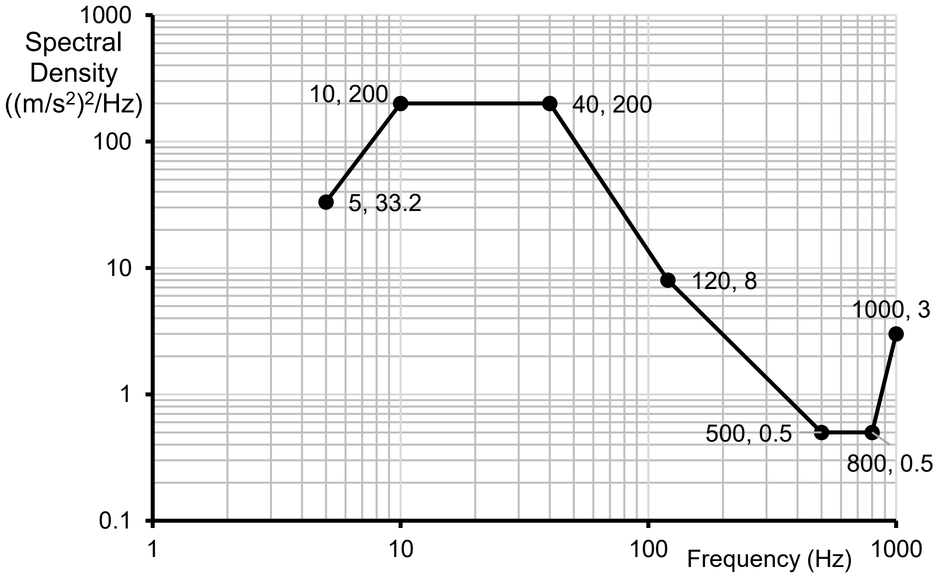


K 15.3.2 Suspension/Axle Mounted (Unsprung Mass)

Random Wide-Band Random Vibration excitation according to IEC 60068-2-64 using the profile in Figure 9.

Test Duration is 8 hours for each axis. RMS acceleration value of 107.3 m/s².

Figure 9 - Vibration Profile – Suspension/Axle mounted (Unsprung Mass)



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K 15.3.3 Engine Mounted

Unless otherwise agreed with the JLR Component Owner the testing shall be completed as a mixed mode vibration test (Sine on Random) using the profiles in Figure 10 and Figure 11 in accordance with IEC 60068-2-80.

A test duration of 22 hours shall be used for each axis.

The random profile shall have an RMS acceleration of 181 m/s^2

Figure 10 - Engine - Random Vibration Profile

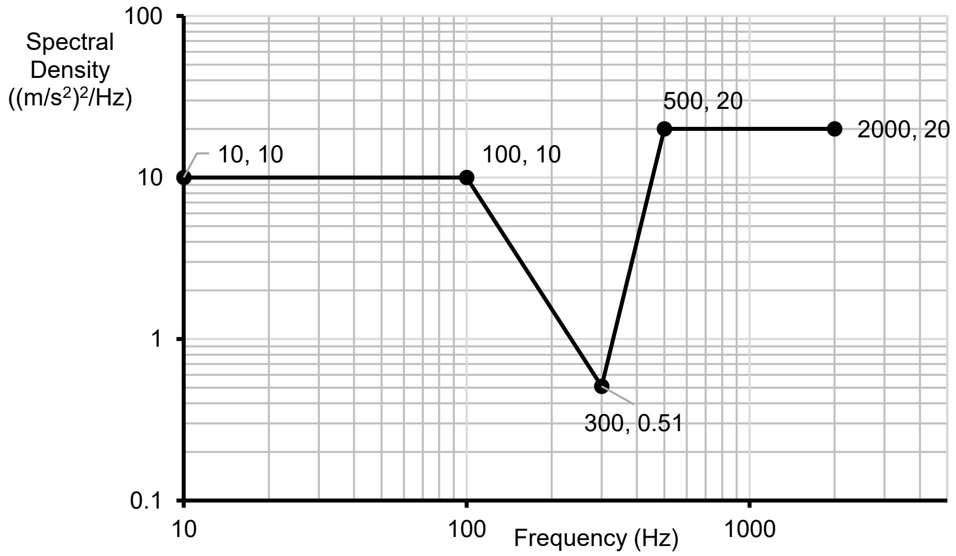
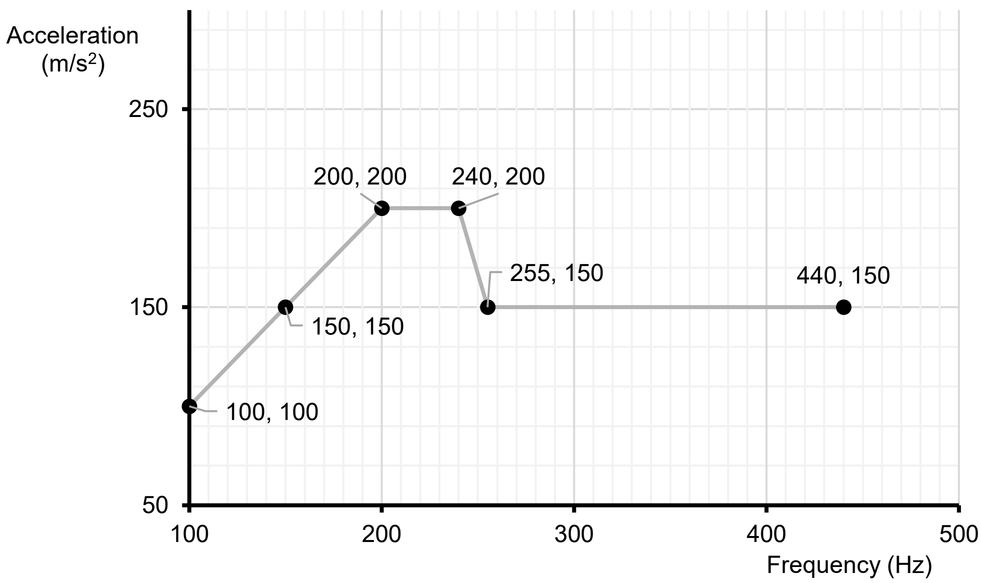


Figure 11 - Engine - Sine Vibration Profile



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K 15.3.4 Gearbox Mounted

Unless otherwise agreed with the JLR component owner, the testing shall be completed as a mixed mode vibration test (Sine on Random) using the profiles in Figure 12 and Figure 13 in accordance with IEC 60068-2-80.

A test duration of 22 hours shall be used for each axis.

The random profile shall have an RMS acceleration of 96.6 m/s^2

Figure 12 - Gearbox Mounted Random Profile

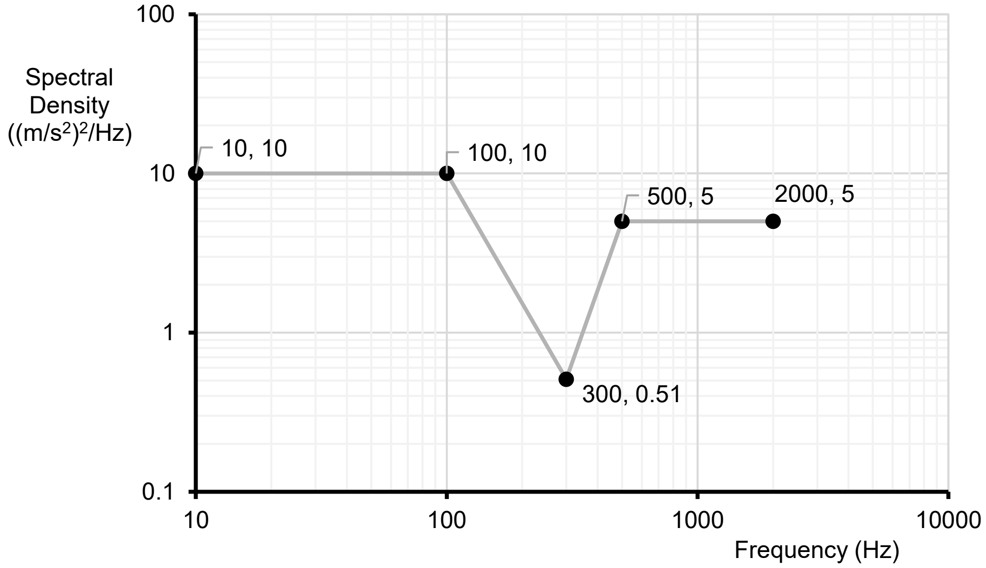
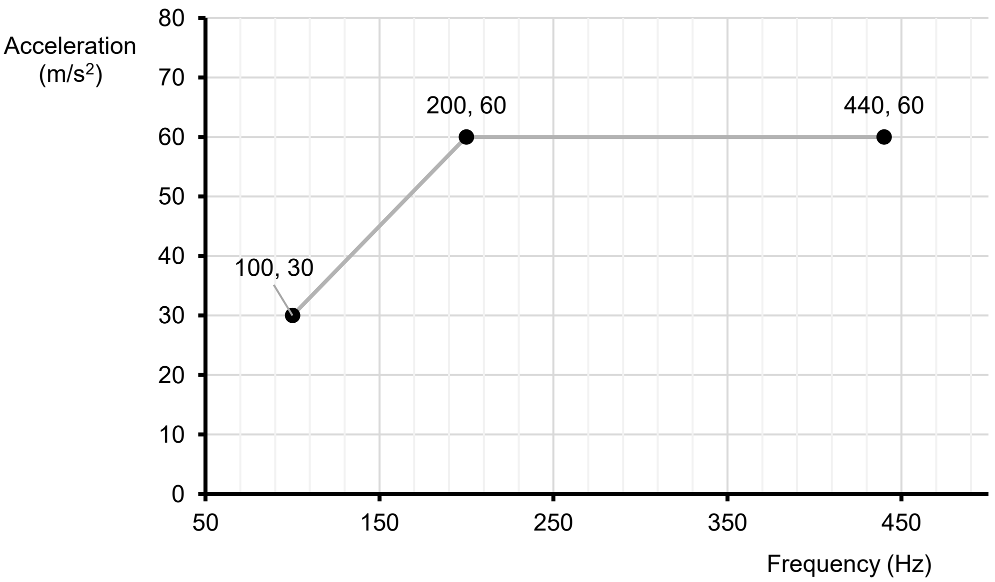


Figure 13 - Gearbox Mounted Sine Profile



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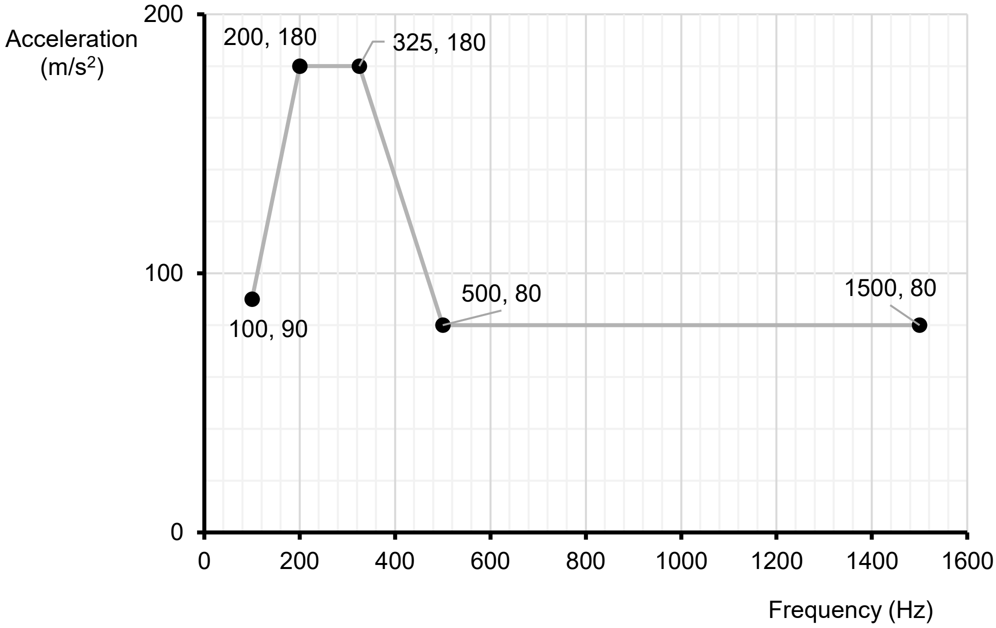
K 15.3.5 Flexible Plenum Chamber

Testing shall be completed in accordance with IEC 60068-2-6 using the profile in Figure 14.

A test duration of 22 hours shall be used for each axis.

Figure 14 - Flexible Plenum Sine Profile



K 15.3.6 Other Locations

Vibration profiles for EE components mounted on vibration inducing parts not covered by the above profiles (such as Pumps, Motors or EDUs, Differential Units or parts mounted on Harnesses or Hoses) shall be agreed between the JLR Component Engineer, D&R PAT Leader/Technical Specialist and the Supplier, based upon measured, simulated or historic vibration data of the device to which it is to be mounted.

Details of the test profile to be used shall be included in the component PDS.

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K 16 **ME-S** - MECHANICAL SHOCK/DROP

These tests are to verify that the component and its shipping container provides sufficient protection from mechanical impact shocks resulting from being dropped during shipping, handling and vehicle production and that the component is capable of withstanding mechanical shocks encountered during vehicle operation.

Possible effects:

- Cracking or breaking of enclosures, mounting points, connectors and mechanical interfaces
- Breaking of parts within the component assembly
- Damage of electrical connections and continuity within the component enclosure
- Malfunction of electro-mechanical devices, mechanisms and controls contained within the component
- Loosening of covers and seals
- Cracking of substrates

Select the applicable mechanical shock/drop tests from Table 4.

Table 4 - Mechanical Shock/Drop Classifications

Classifications			Mechanical Shock/Drop Test Requirement* See Note 3					
No.	Description	Examples of Component Type	K 16.1 Package Drop	K 16.2 Handling Drop	K 16.3 Medium Mech. Shock	K 16.4 Low Mech. Shock	Required Component Labelling	Processing/ Handling Approach
I	Non–robust Component Design/ Careful Handling & Protective Packaging	Instrument clusters, displays, etc.	X (See Note 1)			X	"Set aside if dropped"	Minimize rough handling. Control damaged material.
II	Semi–robust Component Design Large Mass	Electronic modules, large mass etc.	X (See Note 1)		X		"Set aside if dropped"	Minimize rough handling. Control damaged material.
III	Robust Component Design/ Minimal Handling Precautions	Encased sensors, small modules, etc.		X			N/A	No special controls.

Note 1: K 16.1 Package Drop is required only for packages designed for human handling (18 kg max).
Note 2: K 16.1 & K 16.2 require 14 test samples, 1 drop per sample, 6 sides and 8 corners.
Note 3: The High Repetition Mechanical Shock (BUMP TEST) K 16.5 shall be used as defined by the JLR Component Engineer according to the component location.
* For each classification perform the tests identified with an "X".

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K 16.1 Package Drop.

NOTE: Package Drop is required only for packages designed for human handling (18 kg max).

NOTE: It is required that the JLR Component Engineer contact the Dunnage Engineer at the JLR VO plant where this DUT is to be used to determine if there are plant specific Package Drop/Validation tests in which case the plant tests will be run in place of these. If none exist this test or an equivalent test shall be run during PV validation.

K 16.1.1 Equipment and Facilities.

DUT in its protective shipping package and a concrete or steel surface and stanchion to control drop height and orientation.

Ring stand or equivalent.

Quick disconnect hook.

K 16.1.2 Test Setup and Preparation.

Install the DUT and its protective shipping package in the holding fixture.

K 16.1.3 Procedure.

NOTE: The component specification shall define whether one or more packages are to be used for this test. It is suggested that 14 packages be used, one drop per package.

a Drop the package a distance of 1 m onto a concrete or steel surface. Repeat once for each package surface and corner.

b When the test is complete open the package and perform the Functional Check and Visual Check as specified in sections L 2 and L 5.

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K 16.2 Handling Drop.

K 16.2.1 Equipment and Facilities.

DUTs and a concrete or steel surface and stanchion to control drop height and orientation.

Ring stand or equivalent.

Quick disconnect hook.

K 16.2.2 Test Setup and Preparation.

Concrete or steel surface. The drop height is 1 m.

K 16.2.3 Procedure.

NOTE: 14 samples are required for this test.

a Drop once on each face and once on each corner, one DUT per drop.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

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K 16.3 Medium Mechanical Shock.

K 16.3.1 Equipment & Facilities.

Shock machine and necessary holding fixtures must be capable of producing a half sine shock impulse of 981 m/s² (100 g) and 10 ms duration.

K 16.3.2 Test Setup & Preparation.

Install the DUTs on the Shock Machine. It is highly recommended that the vehicle mounting bracket/tabs **NOT** be used to secure the DUT.

K 16.3.3 Procedure.

a Expose the DUTs to 10 ms half-sine shock impulses of 981 m/s² (100 g), one in each opposite direction of each perpendicular axis.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

K 16.4 Low Mechanical Shock.

K 16.4.1 Equipment & Facilities.

Shock machine capable of producing a half sine shock impulse of 491 m/s² (50 g) and 10 ms duration.

Holding fixtures.

K 16.4.2 Test Setup & Preparation.

Install the DUTs on the Shock Machine. It is highly recommended that the vehicle mounting bracket/tabs NOT be used to secure the DUT.

K 16.4.3 Procedure:

a Expose the DUTs to 10 ms half-sine shock impulses of 491 m/s² (50 g), one in each opposite direction of each perpendicular axis.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

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K 16.5 High Repetition Mechanical Shock. (BUMP TEST)

ISO 16750-3 Test 4.2.1

K 16.5.1 Equipment & Facilities.

Shock machine capable of producing half sine shock impulses of 491 m/s² (50 g) and 11 ms duration.

Holding fixtures.

NOTE: It is highly recommended that the vehicle mounting bracket/tabs NOT be used to secure the DUT.

K 16.5.2 Test Setup & Preparation.

Install the DUTs on the Shock Machine. The DUT may be subject to Monitored Operation as specified in L 3 at the discretion of the JLR Component Engineer.

K 16.5.3 Procedure.

a Expose the DUTs to Shock Pulse defined above for the number of shocks according to location shown in Table 3 in each opposite direction of each perpendicular axis.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue to the next test.

Location (on or immediately surrounding)	Number of Shocks
Driver's door, cargo door	13,000
Passenger doors	6,000
Boot lid, tailgate	2,400
Bonnet	720

Table 3 - Table 5 - Bump Test Shocks

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K 17 **AN** - AUDIBLE NOISE

TPJLR.00.187

This test is to verify that the component/bracket assembly will not emit audible noise in the vehicle.

Possible effects are buzzes, squeaks, rattles, etc. which annoy vehicle occupants.

NOTE: The applicability of this test shall be as agreed with the JLR Component Engineer and the NVH PAT leader.

K 17.1 Equipment and Facilities

As TPJLR.00.187 Section G.

K 17.2 Test Setup and Parameters

As TPJLR.00.187 Section H.

Testing profiles shall be obtained from the JLR Noise and Vehicle Harshness (NVH) Programme Attribute Team (PAT) Leader.

K 17.3 Procedure

a Perform the Dynamic Test for components in the interior of the vehicle (Package Zone A - In Passenger Compartment). The DUT shall be mounted rigidly to the shaker table in the orientation that it will be packaged in the vehicle. Where applicable the vehicle mounting strategy shall be used.

b Audio recordings shall be supplied to the JLR NVH PAT leader.

c When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

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K 18 **ME-C** - CONNECTOR AND LEAD/LOCK STRENGTH

These tests are to verify that the component electrical connector, leads and locking mechanisms are of sufficient strength to withstand mechanical stresses (pull, twist etc.) encountered in the vehicle and during assembly, service and maintenance.

NOTE: The Jaguar Land Rover Component Engineer may specify additional tests to verify adequate sealing of moulded connectors.

Possible effects are loss of/intermittent connection.

NOTE: This test series is to verify the module side of the connector system. Failure modes of interest are in the RED areas of Figure 15 and Figure 16.

Figure 15 - Connector Tests - Fly Leads to PCBA

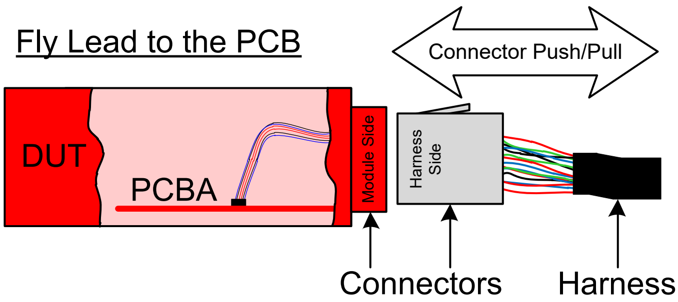
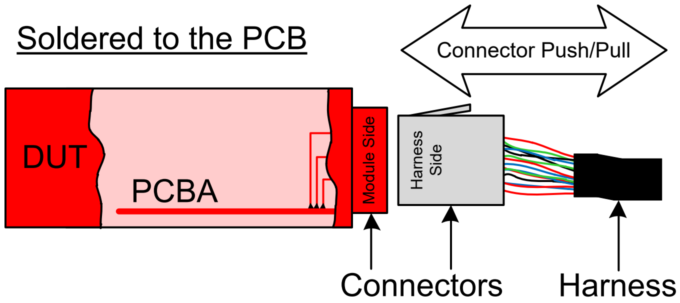


Figure 16 - Connector Tests - Soldered Connection to PCBA



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K 18.1 Connector Push Test.

K 18.1.1 Equipment and Facilities.

Production Connectors and the module/component they attach to.

Holding fixtures and force gauge.

K 18.1.2 Test Setup and Preparation.

Install the DUT and harness assembly in holding fixture.

K 18.1.3 Procedure.

a Apply 1.5 times the Rated Maximum Engagement force to the connector in the direction of mating for one minute.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue to the next test.

Additional Requirement, any visual displacement of the connector during the test shall be agreed with the JLR Component Engineer, Manufacturing and Service.

K 18.2 Connector Torque Test.

K 18.2.1 Equipment and Facilities.

Production Connectors and the module/component they attach to.

Holding fixtures and force gauge.

K 18.2.2 Test Setup and Preparation.

Install the DUT and harness assembly in holding fixture.

K 18.2.3 Procedure.

a Independently apply 39.2 N (4 kgf) to all four sides of the harness connector(s), perpendicular to the direction of mating, at the greatest moment arm of each axis, for one minute.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

Additional Requirement, any visual displacement of the connector during the test shall be agreed with the JLR Component Engineer, Manufacturing and Service.

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K 18.3 Connector Durability Test.

K 18.3.1 Equipment and Facilities.

Fully populated mating connectors with terminals (one connector per component).

Production representative torque tools set to the maximum specified fastener torque.

K 18.3.2 Test Setup and Preparation.

Install the DUT and harness assembly in holding fixture.

K 18.3.3 Procedure.

a Plug and unplug each connection system 11 times by hand or use the maximum specified torque if a bolt is required. The time to mate each connection system by hand must not exceed four seconds.

b When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

K 18.4 Lead/Lock Pull Test.

K 18.4.1 Equipment and Facilities.

Production Connectors and the module/component to which they attach.

Holding fixtures and force gauge.

Size of each wire in connector.

K 18.4.2 Test Setup and Preparation.

Attach the harness connectors to the DUT and install in a holding fixture.

K 18.4.3 Procedure.

a If DUT has individual wire(s) pull each as follows:

- to 0.3 mm²: 49 N
- >0.3 mm² to 0.5 mm²: 78.5 N

b If DUT has a wire bundle pull the wire bundle: 98.1 N

c When the test is complete perform the Functional Check and Visual Check as specified in sections L 2 and L 5. If the DUTs pass continue on to the next test.

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K 19 **CL-SALT** - SALT MIST ATMOSPHERE

IEC 60068-2-11

This test is to verify that the component can withstand the effects of continued exposure to salt water and/or salt atmosphere conditions resulting from vehicle operation on salted roadways and/or in high salt-humidity atmosphere regions such as coastal regions.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-11 must be listed in the test facility's accreditation schedule.

Possible effects:

- Impairment due to salt deposits
- Production of conductive coatings
- Corrosion due to electro-chemical reaction
- Corrosion of metallic surfaces
- Clogging or binding of moving parts.

K 19.1 Equipment and Facilities.

A test chamber with:

a Supporting racks designed and constructed so that they will not affect the characteristics of the salt fog mist. All parts of the test chamber and the supporting racks that come into contact with the DUTs shall be constructed of material or will be buffered with material that will not cause electrolytic corrosion. Condensation is not allowed to drip on the DUTs. No liquid that comes in contact with the DUTs or the test chamber shall return to the salt solution reservoir. The exposure chamber shall be properly vented to prevent pressure build up.

b The capability to maintain temperature in the exposure zone at 35 ±2 °C. The use of immersion heaters within the chamber exposure area to maintain the temperature within the exposure zone is prohibited.

c A salt solution reservoir made of material that is nonreactive with the salt solution, e.g. glass, hard rubber or plastic.

d A means for injecting salt solution into the test chamber. Atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog. Atomization of approximately 2.8 litres of salt solution per 0.28 cubic metre (10 cubic feet) of chamber volume per 24 hours.

e Salt fog collection receptacles placed so that a clean receptacle at any point in the exposure zone will collect from 0.5 to 3.0 millilitres of solution per hour for each 80 square centimetres (10 cm diameter) of horizontal collecting area in an average test of at least 16 hours. A minimum of 2 receptacles shall be used, one placed nearest to any nozzle and one farthest from all nozzles: receptacles shall be placed so that they are not shielded by the test item and will collect no drops of solution from the test item or other sources. Measured results are to be recorded in the test log.

f Equipment required to properly monitor and verify that the DUTs are functioning properly (If required).

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K 19.2 Test Setup And Preparation: Controls

The salt solution shall be heated to 35 ±5 °C before injection into the test section.

Water used during the salt fog test shall be chemically pure with resistance of not less than 500 Ωm.

Air circulation shall be zero.

The atomisers shall be setup so that no direct spray impinges upon the specimens under test.

K 19.3 Preparation For Test:

Salt Solution

The salt used for this test shall be high quality sodium chloride containing not more than 0.1 % sodium iodide and not more than 0.3 % total impurities when dry. A 5 % ±1 % solution shall be prepared by dissolving 5 parts by weight of salt into 95 parts by weight of water. The solution shall be adjusted to, and maintained at, a specific gravity of 1.022 to 1.037. Sodium tetraborate (borax) may be added to the salt solution as a pH stabilization agent in a ratio not to exceed 0.7 g sodium tetraborate to 75 litres of salt solution. The pH of the salt solution, as collected as fallout in the exposure chamber, shall be maintained between 6.5 and 7.2 with the solution temperature at 35 ±5 °C. Only diluted chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust the pH. The pH measurement shall be made electrometrically or colourimetrically.

DUTs:

The test samples shall have Functional and Visual Checks as specified in sections L 2 and L 5.

All surfaces shall be clean and free from oils, greases or other protective materials not part of the design.

Mount the DUTs in the chamber, orientation shall simulate that used in the vehicle.

Connect the monitoring instrumentation and verify that the DUTs are functioning properly (if required).

K 19.4 Procedure.

a Adjust the test chamber temperature to 35 °C ±2 °C. Condition the DUTs for two hours before introducing the salt fog.

b Expose the DUTs to the salt atmosphere for the time specified in the PDS and in section M 4. Continuously atomize a salt solution as specified above for the period specified and during the entire exposure period the salt fog fallout rate and pH of the fallout solution shall be measured at 24 hour intervals.

c When the test is complete the DUTs may be allowed to dry in the ambient atmosphere for 48 h or may have the Functional Check L 2, the Visual Check L 5 and Internal Inspection L 6 performed immediately. If the DUTs pass continue on to the next test.

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d During the visual examination of the DUTs pay particular attention to:-

- High stress areas.
- Areas where dissimilar metals are in contact.
- Electrical and electronic DUTs - especially those having closely spaced, unpainted, or exposed circuitry.
- Metallic surfaces.
- Enclosed volume where condensation has occurred or may occur.
- DUTs or surfaces provided with coating or surface treatments for corrosion protection.
- Cathodic protection systems:- mechanical systems subject to malfunction if clogged or coated with salt deposits.
- Electrical and thermal insulators.

K 20 **CL-SOL** - SOLAR RADIATION

The validation of a part being robust against solar loading will typically be covered by the careful selection of materials. Where a physical validation is required the following test method shall be used.

IEC 60068-2-5

This test is to verify that components mounted in areas exposed to the sun can withstand prolonged periods of solar radiation.

Possible effects:

- Degradation of materials (cracking, softening, warping, embrittlement etc.)
- Damage to optical sensors

Levels of environmental stress shall be agreed between the Component Engineer the supplier and the TASE PAT leader and shall be based upon vehicle and/or glasshouse simulation data.

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K 21 **CL-CG** - CORROSIVE GASES

IEC 60068-2-60 Test Ke Method 4

This test is to verify that the component can withstand the effects of common atmospheric pollutants that can occur in high concentrations in built-up areas and/or industrial areas.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-60 Test Ke must be listed in the test facility's accreditation schedule.

Possible effects:

- Corrosion of exposed metal surfaces
- Degradation of switch contacts

NOTE: Where all of the PCBA components used within the DUT are automotive qualified, including AEC Q102 for all optoelectronic components evidence of this may be presented in lieu of this test, with the exception of switchgear.

K 21.1 Equipment and Facilities

Test equipment as specified in IEC 60068-2-60

Equipment required to validate the operation of the DUT

K 21.2 Test Procedure and Setup

Concentration of Corrosive gases shall be measured in the test environment and where possible shall be subject to closed loop control. Actual concentrations achieved shall be included in the test report.

K 21.3 Procedure

a Complete the test according to IEC 60068-2-60 Test Ke Method 4 with the following parameters:
Duration: 21 days
Gas Concentration: Table 6

b Once the test is complete perform a Functional Check and the Visual Check as per sections L 2 and L 5

Table 6 - Corrosive Gases and Concentrations

Gas	Name	Concentration Parts per Billion by Volume
H_2S	Hydrogen Sulphide	10 ±5
NO_2	Nitrogen Dioxide	200 ±20
Cl_2	Chlorine	10 ±5
SO_2	Sulphur Dioxide	200 ±20

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K 22 **CR** - CHEMICAL RESISTANCE

ISO 16750-5

This test is to verify that the component can withstand the effects of chemicals (cleaning compounds, lubricants, automotive fluids, etc.) encountered during component manufacture, vehicle assembly, service and maintenance, and during vehicle use.

Possible effects:

- Degradation of materials (cracking, softening, warping, etc.)
- Deterioration of label adhesives
- Discolouring of materials
- Smudging of inks and dyes.

NOTE: Where the materials used have previously been validated for resistance against specific chemicals, these chemicals may be omitted.

K 22.1 Equipment and Facilities.

Ample amounts of each fluid to coat the DUT.

A test chamber capable of meeting the temperatures stated in M 5.

K 22.2 Test Setup and Preparation.

Determine the test chemicals and parameters from the PDS.

K 22.3 Procedure.

a Apply the specified chemical solutions onto exterior surfaces of the DUTs using the agreed application method(s).

b Install the DUTs in the chamber. Power up the chamber and stabilize at the specified temperature.

c Expose the DUTs for the specified duration.

d When the test is complete perform the Functional Check (L 20) and Visual Check (L 5). If the DUTs pass continue to the next test.

NOTE: It is suggested that one sample per fluid be used.

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K 23 **L-TSE** - THERMAL SHOCK ENDURANCE

This test is to verify that the component and its PCBA(s) are free from failure during vehicle useful life caused by thermally induced stresses in the vehicle.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-14 test Na must be listed in the test facility's accreditation schedule.

Possible effects are material fatigue due contraction and expansion stresses that occur in:

- Solder
- Solder connections
- PCB traces
- Hybrid circuitry
- Conformal Coating / Solder Resist.

Additionally the growth of metallic whiskers.

NOTE: The aim of this test is to detect mechanical weaknesses of the electrical design of the part, such as, cracking of solder joints, conformal coating or solder resist, delamination of materials etc. which could occur over the lifetime of the part.

K 23.1 Equipment and Facilities.

The chamber(s) shall be capable of maintaining the specified temperature(s) in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.

The thermal shock system shall consist of two thermal chambers with an automated mechanical system to move the test samples from one chamber to the other. Alternatively, a chamber capable of transitioning from T_{max} to T_{min} and vice versa in under 30 seconds.

K 23.2 Test Setup and Preparation.

NOTE: Before start of test, submit a copy of the DUT assembly solder profiles (Reflow, Wave, and Selective Solder) to the JLR Component Engineer with written verification that the profiles meet the solder and component manufacturers' thermal requirements.

NOTE: Where Conformal Coating is used, submit a copy of the curing profile to the JLR Component Engineer (Including UV curable coatings) and written verification this meets the manufacturers' requirements. Additionally, verification that the PCBAs meet the Conformal Coating manufacture's requirements for cleanliness.

NOTE: This test shall be completed on the PCBA(s)

Before the start of test inspect all 8 samples and section 2 of them, for larger or more complex boards (with >20 components as a guide) additional examination by X-ray is preferred.

The solder quality must meet "Acceptable Class 3" according to IPC-A-610 for all sections applicable to the PCBA, including no evidence of large solder voids (30% or greater F 8.3.12.4). An 8D will be started to determine the root cause of non-compliances, this test shall not be performed until compliance is met. Provide pictures of all cross sections and a representative number of each joint type.

The DUTs shall be mounted in the chamber at the same orientation as used in the vehicle. It is highly recommended that the DUTs be rack-mounted, not just placed on the floor of the chamber.

Two of the DUTs shall be thermo-coupled and a few pre-test cycles run to verify that the substrate temperature reaches set point before the start of the test. Put copies of the recorded temperature profiles in the test log.

K 23.3 Procedure.

a Bring both chambers to their respective set temperatures T_{Emin} and T_{Emax} as indicated in

b Figure 17 and start the first cold soak cycle.

c When the DUTs have soaked for t_{dwell} (30 minutes) remove them from the cold chamber and install them in the hot chamber. This must be done within 30 seconds.

NOTE: If the PCBA has been instrumented with thermocouples to measure the surface temperature, the time taken for the PCBA surface to reach 3 K of the set temperature plus an additional 5 mins may be taken as the dwell time.

d Soak the DUTs for t_{dwell} (30 minutes), remove them and install in the cold chamber within 30 seconds. Soak for t_{dwell} (30 minutes) and then remove them from the cold chamber and install them in the hot chamber within 30 seconds.

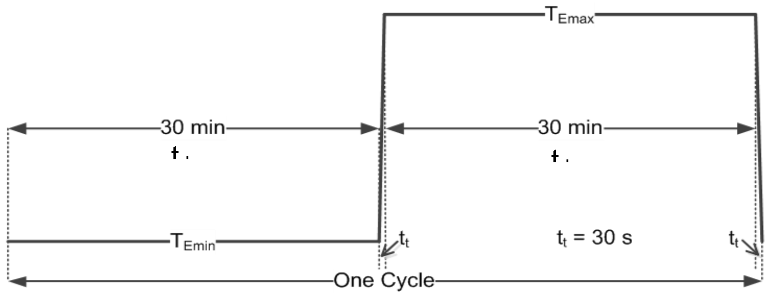
e Continue to cycle the DUTs for 1,000 cycles, performing a Visual Inspection of the joints and coatings at 320±10, 640 ±10 and 1,000 cycles.

f When the test is complete return DUTs to room temperature.

g Perform a Performance Evaluation (L 1) and Visual Check (L 5)

h Additional acceptance requirement: no cracks in the solder, no tin whiskers and no cracks or bubbles in coatings allowed. Three of the samples shall have solder joints examined by sectioning and if used previously x-ray. Cracks that interrupt no more than 30% of the solder joint may be accepted at the discretion of the JLR Component Engineer.

Figure 17 - Thermal Shock Endurance



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K 24 **L-HTE** - HIGH TEMPERATURE ENDURANCE

NOTE: A complete WCCA may be substituted for the High Temperature Endurance test where all high temperature failure modes identified in the D/PFMEA are circuit related, and covered by the WCCA and the component must be QM rated according to ISO 26262. Written agreement with the JLR component owner shall be sought.

NOTE: For components that are operational during EV charging, the parameters of this test shall be determined using the Arrhenius model and the mission profile of the DUT. A copy of JLR-TMP-561681 shall be completed by the supplier and submitted to JLR for approval.

IEC 60068-2-2 (Test Be)

This test is to demonstrate that component electronics are free from time-temperature dependent failure during the intended life of the component.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-2 must be listed in the test facility’s accreditation schedule.

Possible effects:

- Dielectric breakdown
- Damage to Electrolytic Capacitors, Flash memory or other components subject to high temperature wear
- Metal migration
- Intermetallic growth

K 24.1 Electro-migrationEquipment and Facilities.

- The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I. The velocity of air circulation shall be determined from the requirements of IEC 60068-2-2. Additional requirements regarding the airflow may be prescribed in the PDS.
- Equipment required to properly monitor and verify that the DUTs are functioning properly.

K 24.2 Test Setup and Preparation.

NOTE: Careful consideration shall be given to DUTs with liquid cooling or other similar technologies to ensure condensation does not form on the DUT during this test

- Mount the DUTs in the chamber at the same orientation as used in the vehicle.
- Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

K 24.3 Procedure.

a Power up chamber and stabilize at the specified temperature T_{max}/T_{pmax}. Expose the DUTs for 1,000 hours whilst performing Monitored Operation as specified in section L 3. Check the DUTs once per day to verify that they are operating properly.

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b When the test is complete perform the Functional and Visual Check as specified in sections L 2 and L 5.

K 25 **L-CD** - CONTROLS DURABILITY

This test is to demonstrate that component driver/passenger operated controls have no wear out mechanisms which can occur during the intended useful life of the component in the vehicle.

Possible effects:

- Contact wear
- Fatigue of plastic and metal parts
- Electrical intermittent operation/noise
- Mechanical wear
- Binding or sticking.

K 25.1 Equipment and Facilities.

- Holding fixtures and actuators for controls.
- Equipment required to properly monitor and verify that the DUTs are functioning properly.

K 25.2 Test Setup and Preparation.

a Mount the DUTs in the chamber at the same orientation as used in the vehicle. Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

K 25.3 Procedure.

a Operate the driver/passenger controls for the number of cycles specified as the lifetime in the component PDS.

b A temperature distribution shall be selected by the Supplier and JLR Component Engineer for the operation cycles. The following is provided as a guide:

2/3 of the cycles at T_N

1/6 of the cycles at T_{min}/T_{pmin}

1/6 of the cycles at T_{max}/T_{pmax}

c When the test is complete perform the Functional and Visual Check as specified in sections L 2 and L 5.

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K 26 **L-MW** - MECHANICAL WEAR-OUT

This test is to demonstrate that mechanical or electro-mechanical (moving) parts within the component have no wear-out mechanisms which occur during the intended useful life of the component in the vehicle.

Possible effects:

- Contact wear
- Fatigue of plastic and metal parts
- Audible noise
- Mechanical wear
- Binding or irregular operation.

K 26.1 Equipment and Facilities.

- Holding fixtures and actuators for controls.
- The chamber shall be capable of meeting the requirements of section I and of maintaining the specified temperature in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.
- Equipment required to properly monitor and verify that the DUTs are functioning properly.

K 26.2 Test Setup and Preparation.

a Mount the DUTs in the chamber at the same orientation as used in the vehicle.

b Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

K 26.3 Procedure.

a Operate the Electromechanical system (Motor, Gearbox, Solenoid etc) for its specified lifetime duration and/or number of operational cycles.

b Operate:

2/3 of the cycles/duration at T_N

1/6 of the cycles/duration at T_{min}/T_{pmin}

1/6 of the cycles/duration at T_{max}/T_{pmax}

The temperatures and the number of cycles / durations is as specified in the PDS.

c When the test is complete return DUTs to room temperature and perform the Functional and Visual Check as specified in sections L 2 and L 5.

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K 27 **L-THB** - 85/85 HIGH TEMPERATURE - HIGH HUMIDITY ENDURANCE

This test is to demonstrate that the component electronics are free from high temperature-high humidity dependent failures during the intended useful life in the vehicle application.

This test shall be performed in test facilities accredited to ISO 17025 by the appropriate national body, as a minimum testing to IEC 60068-2-78 must be listed in the test facility's accreditation schedule.

Possible effects:

- Swelling of hydroscopic materials
- Electrolytic corrosion of die metallization
- Degradation in electronic materials/circuit performance
- Current leakage
- Metallic whisker growth
- Delamination of Conformal Coating
- Hydrolysing of Conformal Coating
- Dendritic growth

NOTE: Supplier shall provide evidence that all components used in their DUTs meet the 1,000 hour 85 degree Celsius, 85 % relative humidity requirement. Components that do not meet this requirement may not be used in a JLR product.

Once the above requirement is confirmed this test is optional. This test may be used when the failure mode of interest cannot be verified by one of the other tests in this specification.

For electro-mechanical components, e.g. CD players, LCDs, the test time may be adjusted based on industry practice.

K 27.1 Equipment and Facilities.

- The chamber shall be capable of maintaining the specified temperature and humidity levels in the working space within the tolerances given in section H 1.3 and section I. Forced air circulation may be used to maintain homogeneous conditions.
- Supporting racks designed and constructed so that they will not affect the characteristics of the fog mist. All parts of the test chamber and the supporting racks that come into contact with the DUTs shall be constructed of material or will be buffered with material that will not cause electrolytic corrosion. Condensation will not be allowed to drip on or form on DUTs.
- Equipment required to properly monitor and verify that the DUTs are functioning properly.

K 27.2 Test Setup and Preparation.

a Mount the DUTs in the chamber at the same orientation as used in the vehicle.

b Connect the monitoring instrumentation and verify that the DUTs are functioning properly.

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NOTE: Water used during the test shall have a resistivity of not less than 500 Ωm.

K 27.3 Procedure.

a Power up chamber and stabilize at 85 °C / 85 % Relative Humidity and operate the DUTs for 1,000 hours or as determined by the JLR Component Engineer.

b Check the DUTs once per day to verify that they are operating properly.

c When the test is complete return DUTs to room temperature and perform the Functional and Visual Check as specified in sections L 2 and L 5.

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K 28 HALT

NOTE: HALT (Highly Accelerated Life Test) testing is used to discover the operational and if time and samples allow destruct levels for temperature, vibration, thermal shock rate of change and combinations of the same. The results are for information only. It is assumed before starting that the design has been shown to be capable of meeting the standard environmental requirements in EC-7128/TPJLR.18.125. The tests require a special test chamber.

Each non-compliance, either non-operation or destruction, shall be root-cause analysed and possible solutions be investigated. In the past it has been found that for a minimal cost, and occasionally without cost, a failure can be eliminated increasing the robustness of the design.

It is also required that these tests be repeated on the PV samples to look for changes from the DV results. If it is determined that the PV results are acceptable it is then suggested the HALT be added as a IP test requirement, possibly once per month or at a minimum once per quarter for the first year of production and twice a year after that if results indicate process robustness is stable.

A change in results, which usually indicates a weakening of the design as processed, is significant but not a show stopper since the in-op levels should be greater than the spec by a significant margin. This usually indicates a change in the parts or the process that has resulted in a loss of robustness. It is required that the root cause be found and if feasible a fix be implemented to return the process to its previous level.

K 28.1 Equipment and Facilities.

NOTE: It is required that a HALT type chamber be used for this testing capable of temperatures from -100 °C to +200 °C, thermal shock rates from 1 °C to 100 °C/min and vibration excitation of 700 m/s² multi-axis random as a minimum.

- Equipment required to properly monitor and verify that the DUTs are functioning properly during the test.

K 28.2 Test Setup and Preparation.

a It is preferable to test DUTs with the top removed and with two thermo-couples attached to the board surface to monitor temperature.

b Conduct the Performance Evaluation test, L 1, on the DUTs before starting the test and on surviving DUTs at the end of testing.

K 28.3 Procedure.

Use three samples for the temperature and thermal shock tests.

Operational limit tests.

Cold test.

Install 3 DUTs in the chamber, if practical, verify correct operation and attach two thermo-couples to the board surface of one of the samples and use them as controls for the chamber during the test. Reduce temperature to -40 °C at a maximum rate of 4 °C/min. Power up module and verify proper operation. Power off and lower the temp in 5 °C increments with a 5 minute dwell. After each dwell power up and verify proper operation. Continue until improper operation is found or -100 °C is reached.

If a module fails, record the temperature and the improper operation.

Cycle power on and off to see if module will recover. If it recovers dwell for 5 minutes and then continue to the next step. If not power off and return to the temp where the module properly performed, dwell 5 minutes, power up and verify proper operation. If module doesn't recover, power off and continue the 5 °C increase steps with 5 minute dwell until proper operation returns or -40 °C is reached. If the module recovers note the temperature at which it recovers and perform the functional check at the voltages specified by the JLR Component Engineer.

When complete return the module to room temperature at 4 °C/min.

If module doesn't recover note this in the test log and remove for root-cause analysis at a later date. If not all samples failed return to the previous set point at 4 °C/minute and continue until all have failed or reached -100 °C. Return to ambient when complete at 4 °C/min and move to the Hot test. If the samples return to proper operation at -40 °C or lower they may be carried over to the Hot test. If not replace them with new modules and move on.

NOTE: Loss of operation above -55 °C is considered a failure that must be 8D analysed and fixed.

Hot test.

Install 3 DUTs in the chamber if practical, verify correct operation and attach two thermo-couples to the board surface of one of the samples and use them as controls for the chamber during the test. Raise temperature to +75 °C at 4 °C/min max. Power up module and verify proper operation. Power off and raise the temp in 5 °C increments with a 5 minute dwell. After each dwell power up and verify proper operation. Continue until improper operation is found or until T_{max} +50 °C is reached.

If a module fails, record the temperature and the improper operation.

Cycle power on and off to see if module will recover. If it recovers dwell for 5 minutes and then continue to the next step. If not power off and return to the temp where the module properly performed, dwell 5 minutes, power up and verify proper operation. If module doesn't recover, power off and continue the 5 °C decrease steps with 5 minute dwell until proper operation returns or +75 °C is reached. If the module recovers note the temperature at which it recovers and perform the functional check at the voltages specified by the JLR Component Engineer.

When complete return the module to room temperature at 4 °C/min.

If module doesn't recover note this in the test log and return module to room temp at 4 °C/min max and remove for root-cause analysis at a later date. If not all samples failed return to the previous set point at 4 °C/min and continue until all have failed or reached T_{max}/T_{pmax} +50 °C. Return to ambient when complete at 4 °C/min and move to the Thermal Shock test. Samples that return to proper operation at +75 °C or higher they may be carried over to the Thermal Shock test. If not replace them with new modules and move on.

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NOTE: Loss of operation below T_{max} +15 °C is considered a failure that must be 8D analysed and fixed.

Thermal Shock test.

Install the DUTs in the chamber, verify correct operation and attach two thermo-couples to the board surface of one of the samples and use them as controls for the chamber during the test.

Reduce temperature to -40 °C at 4 °C/min. max. Power up module and verify proper operation. Start the thermal shock test with a ramp rate of 15 °C/min to T6 and dwell for as long as it takes the board to come to temperature and then power off the module and return to -40 °C. Repeat this temp/power on & off cycle 5 times.

Increase the ramp rate in 5 °C/min increments and repeat the 5 cycles until a module fails or the ramp rate equals 50 °C/min.

If a module fails, record the ramp rate and the improper operation. Power the module off and on to see if it will recover. If yes make a note in the test log and continue to the next step until 50 °C/min is reached or another failure point is reached which doesn't recover with power cycling. Note ramp rate and the improper operation power off the module and return the ramp rate to the level where the module last operated properly. Power up the module after 1 cycle and verify operation level. If it fails to recover continue to lower the ramp rate until the module recovers or 10 °C/min is reached. If the module recovers note the ramp rate in the test log. Return the DUT to ambient temperature at a max of 4 °C/min and perform the functional check at V_N.

If module doesn't recover note this in the test log and remove the DUT for root-cause analysis at a later date. If not all samples failed return to the previous ramp rate and continue until all have failed or you've reached 50 °C/min. Return to ambient when complete and move to the Vibration test.

NOTE: Loss of operation at a ramp rate less than 30 °C/min is considered a failure that must be 8D analysed and fixed if the mini validation test methods are to be used.

Vibration test. Three new samples may be used for this test.

Firmly mount cased modules to table and mount a tri-axial accelerometer to the case of one of the samples and if practical mount two thermo-couples to the board surface. Control the vibration level using the accelerometer mounted on the case if possible.

Power up and verify proper operation.

Start vibration, continue to monitor the module and slowly ramp up to 5.0 g. Dwell there for 5 minutes. If module continues to operate properly increase the vibration in 5.0 g increments with 5 minute dwells until a module fails to operate properly or 20 g is reached. After 20 g is reached increment in steps of 2.5 g until 50 g is reached or the module fails. If module fails, record the vibration level and the improper operation and cycle power on and off to see if module will recover. If it recovers continue to vibrate at this level for 5 minutes and then go to the next step. If not power down module and return to the last

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vibration level at which the module operated properly and power up and check for proper operation. If it doesn't recover power off and decrease vibration in 5.0 g increments with 2 minute dwells until operation returns or 5.0 g is reached. If the module recovers note the g level and perform the functional check at the various voltages as defined by the JLR Component Engineer. If module doesn't recover note this in the test log and return module to room temp at 4 °C/min max and remove for root-cause analysis at a later date. If some samples did not fail return to the previous set point and continue until all have failed or reached 50 g. If the samples survive they should be used for the Thermal Shock/Vibration test. If not replace them with new modules and move on.

Where a module does not recover note this in the test log and remove for root-cause analysis at a later date.

NOTE: Loss of operation below 20g is considered a failure that must be 8D analysed and fixed.

Thermal Shock/Vibration test.

Verify that the DUTs are securely mounted to the table. Power up and verify proper operation.

Reduce temperature to -40 °C at 4 °C/min. Start the thermal shock cycle at the ramp rate the modules last performed properly during the Thermal Shock test. Start vibration at the g level the module last operated properly at the same time. Turn the vibration on & off in 2 minute intervals and turn the power off when the temperature ramps down and back on when the temperature ramps up.

Use the thermo-couples mounted to the board to verify the board is at the min or max temperature before the return ramp cycle begins.

Continue for 20 cycles or until the modules perform improperly. If a module fails note in the test log the improper operation and whether the vibration was on or off, temperature state, increasing, decreasing or stable.

Power off the module and turn back on to see if it recovers, if yes continue test, if not power off and turn off the vibration if it is on and return the module to room temperature at 4 °C/min. Power-up the module to see if it recovers if yes, note it in the test log and remove the module.

If there are modules which were still operating properly return to point of the previous failure and continue to 20 cycles or the next failure.

If the modules survive 20 cycles note this in the test log and turn off the vibration and return the modules to room temperature at 4 °C/min.

All surviving modules shall run through the Performance Evaluation test, L 1, and the results shall be compared to those taken in the pre-test evaluations.

End of test

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L ACCEPTANCE TESTS

L 1 PERFORMANCE EVALUATION

To verify that the component meets electrical and mechanical performance requirements at the limits of the component's performance temperature / voltage specifications

NOTE: Significant performance targets for the module shall be selected for measurement by the JLR component engineer.

Select component performance parameters with quantifiable acceptance criteria/limits (e.g. voltages, currents, frequencies, input/output characteristics, display/illumination intensities, CIE Chromaticity Coordinates, computations, KOL (key off load) and time-outs).

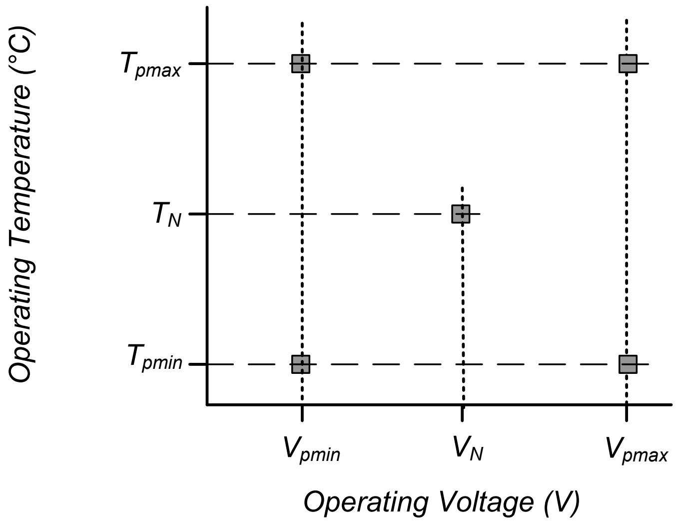
L 1.1 Method

a Minimum, maximum and nominal test supply voltages (V_{pmin}, V_{pmax}, and V_N respectively) shall be according to the Components PDS, section M 1 is provided for guidance and reference.

b Minimum, Maximum and Nominal Temperatures (T_{min}/T_{pmin}, T_{max}/T_{pmax} and T_N respectively) for performance evaluations. Temperatures shall be selected in line with the component PDS. Temperature selection should be in accordance with M 2.

Measure and record component performance parameters specified in the component PDS at the 5 temperature-voltage points shown in Figure 18.

Figure 18 - Performance Evaluation Temperatures and Voltages



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L 2 FUNCTIONAL CHECK

To reasonably verify component function and performance between individual environmental tests.

L 2.1 Method

a Operate the component at Function Check Test Voltage V_N (as per section M 1) and specified load stress conditions at either room temperature T_N or test chamber set temperature.

b Evaluate a subset of the component function/performance characteristics specified in the PDS. Record any discrepancies or abnormal operation.

c Evaluate any faults reported by the DUT completing an 8D for each.

d Functional Checks shall be performed immediately after or within 24 hours following each environmental test as appropriate and/or as specified.

L 2.2 Acceptance

Identify the characteristics necessary to verify satisfactory function after each environmental stress. Qualitative requirements for the characteristics shall be defined.

L 3 MONITORED OPERATION

To verify that the component operates properly in the environment in which it is being evaluated.

L 3.1 Method

a Operate the component at Monitored Operation Test Voltage V_N as per section M 1 and specified load stress conditions during the environmental test.

Maximum Load Stress: Maximum realistic load applied to the DUT to maximise self-heating

Normal Load Stress: Typical loading applied to the DUT in line with a typical drive cycle scenario

b Monitor component characteristics to verify proper operation as specified in the component PDS. Record any discrepancies or abnormal operation.

L 3.2 Acceptance

Functional or performance characteristics with quantifiable criteria/limits necessary to determine satisfactory operation during exposure to the environments.

NOTE: Characteristics should be determined based on subsystem operational requirements, functional importance class, FMEA's, applicable regulatory/safety requirements, company standards and the "Typical Environmental Effects" listed in both section M 6 and at the head of each test as applicable.

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L 4 FUNCTIONAL EVALUATION

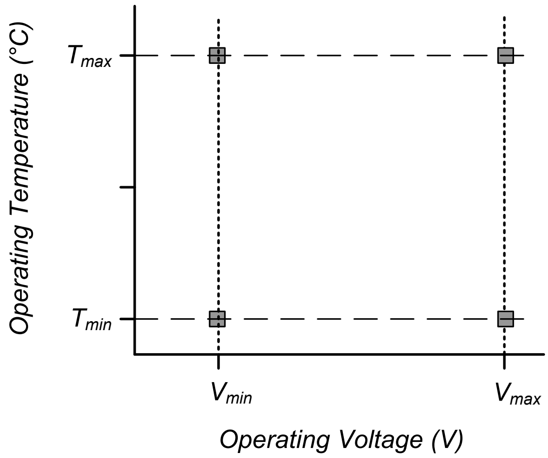
To verify that the component meets functional requirements at temperatures and/or voltages beyond specified performance requirements.

NOTE: The temperatures specified shall not exceed the maximum and minimum storage temperatures.

L 4.1 Method

Evaluate component functional characteristics specified in the Component PDS at the 4 temperature/voltage points shown in Figure 19 if different than those required in L 1. Test voltages (V_{min} and V_{max}) and test temperatures (T_{min}, T_{max} and T_{pmax}) per sections M 1 and M 2 respectively. Record observations.

Figure 19 - Functional Evaluation - Temperatures and Voltages



L 4.2 Acceptance

Meet required Functional Characteristics.

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L 5 VISUAL CHECK

To inspect the component for possible damage that may have occurred during an environmental test.

L 5.1 Method

a Examine the component for visible signs of deterioration or damage.

b Photograph the component from all applicable angles. Photographs shall be retained and may be included in test reports if applicable

c Photographs of DUTs or other applicable parts shall be to an acceptable standard (see IPC-A-610 Revision F Section 1.8, 1.10 and 1.11)

d Dimensional measurements shall be performed to establish acceptability if requested in the either component PDS or by the JLR component engineer.

e Visual Checks shall be performed immediately after or 24 hours following each environmental test as appropriate and/or as specified.

L 5.2 Acceptance

Using a comprehensive listing of visual characteristics with sufficient qualitative criteria to determine functionality or degradation resulting from exposure to the environments.

NOTE: The characteristics and criteria should be based on vehicle requirements, subsystems requirements, corporate standards, and "Typical Environmental Effects" listed in section M 6 and at the head of each test as applicable.

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L 6 INTERNAL INSPECTION

To assess the condition of the PCBA, electrical connections, internal mechanisms, etc.. Typically performed at the end of a test group.

L 6.1 Method

a Using appropriate physical and/or visual techniques inspect PCBAs and other internal parts to evaluate the condition of electrical interfaces and connections.

b Where ingress tests have been completed, using appropriate techniques, inspect the PCBA for any signs of dust or moisture ingress. Special attention shall be paid to breathable membranes where these are employed in the design.

c Internal inspections shall be performed, when required, within 24 hours following the environmental test as appropriate and/or as specified. Inspection procedures are not sensitive to time elements and/or requiring scheduling of laboratory facilities may be performed at extended time periods.

d Forces and Torques required for disassembly may be measured at the request of the Component Engineer

e Photography of all PCBAs or other parts examined as part of the acceptance procedure. Photographs shall be to an acceptable standard (see IPC-A-610 Revision F Section 1.8, 1.10 and 1.11)

L 6.2 Acceptance

Using a comprehensive listing of internal visual characteristics or physical analysis, with sufficient qualitative criteria that can determine degradation of interconnections and function. Specific criteria should include “Typical Environmental Effects” from M 6 as applicable.

Suggested methods of analysis

- Photography
- Light Microscopy
- SEM including EDX
- Solder joint cross sectioning
- X-Ray analysis

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M GENERAL INSTRUCTIONS

M 1 VEHICLE SUPPLY VOLTAGE RANGES

NOTE: The following tables do not consider voltage drops due to vehicle harnesses. This shall be taken into consideration for all component voltage requirements

	Function Performed with Engine Running		**Functions Performed with Ignition Off (Battery Supply Only)**	
Component Test Voltage	**Requirement for Component in the Vehicle Application**	**Voltage (V_{DC}) (See Note 1)**	**Requirement for Component in the Vehicle Application**	**Voltage (V_{DC}) (See Note 1)**
V_{min} Low Voltage of Guaranteed Function	1 Must remain functional in a stalled or a near–stalled engine idle condition.	6	1 Must remain functional when battery is in a low state of charge	9 or 10
	*2. Must remain functional during all engine idle conditions including when vehicle electrical load exceeds charging system capacity.	8 or 9	2 Loss of function is acceptable when battery is in a low state of charge or under heavy electrical load.	11
	3 Loss of function is acceptable under condition described in (1) and (2) above.	10		
V_{max} High Voltage of Guaranteed Function	1. Must function during engine warm–up period after cold start (-40 °C to -30 °C).	15.5 or 16	1 Must remain functional when a battery charger is connected to the charging system.	15 or 16
	2 Function not required during engine warm–up period after cold start (-40 °C to -30 °C).	15 or 15.5	2 Loss of function is acceptable when a battery charger is connected to the charging system.	14
V_{pmin} Low Voltage of Guaranteed Performance	1. Must perform to specification in a stalled and/or near–stalled engine condition.	6 or 7	1 Must perform to specification when the battery is in a low state of charge.	10 or 11
	*2. Must perform to specification during all idling conditions (including cases when electrical load exceeds charging system capacity.)	8, 9 or 10	2 Performance to specification is not required when the battery is in a low state of charge.	11
	3. Performance to specification not required under conditions. (1) and (2) above.	10, 11 or 12		
V_{pmax} High Voltage of Guaranteed Performance	1. Must perform to specification during engine and warm–up period after cold start (-40 °C to -30 °C).	15, 15.5 or 16	1. Must perform to specification when a battery charger is connected to the vehicle charging system.	15 or 16
	2. Performance to specification not required during warm–up period after cold start (-40 °C to -30 °C).	14.5 or 15	3. Performance to specification is not required when a battery charger is connected to the vehicle charging system.	13 or 14
V_N Nominal Operating Voltage	Same for all components.	13.5	Same for all components.	12.5

Note 1: Select the most appropriate component test voltage requirement (for V_1, V_2, V_3, V_4 and V_N) from the table while also considering the following items:

a. Component and subsystem Function Safety Classification and Functional Performance Classification.
b. Subsystem configuration and component requirements for orderly shut-down at low voltage, fault tolerance, etc.
c. The charging system performance characteristics for all vehicle applications and operating conditions (also see Table 8).
d. Voltage drops which may be expected in wiring/distribution systems in **all** vehicle applications.
e. Value selected for V_3 must be $\geq V_1$.
f. Value selected for V_4 must be $\leq V_2$.
g. Subsystems/components required to operate when the engine is running and not running must meet both requirements; i.e., Voltages must be specified considering both sides of the table appropriately.

Note 2: V_1 to V_4, V_N voltage tolerance is ± 0.2V.

Note 3: Functional Check Test Voltage and Monitored Operation Test Voltage V_N ± 0.5V.

*This condition is considered normal and can be expected to occur occasionally in any vehicle.

Table 7 - Vehicle Supply Voltage Ranges - Functions

Table 8 - Vehicle Supply Voltage Ranges - Operating Conditions

<table>
<tr><th colspan="5">Description of Vehicle Conditions</th></tr>
<tr><th colspan="2">Ignition–On/Engine Running</th><th rowspan="2">System Voltage (VDC) Measured at Battery*</th><th colspan="2">Engine Not Running (Battery supply only)</th></tr>
<tr><th>Vehicle Operating Condition</th><th>Frequency of Occurrence</th><th>Vehicle Operating Condition</th><th>Frequency of Occurrence</th></tr>
<tr><td>Highest voltage regulation occurs during engine warm–up periods following cold starts at extreme low temperatures.</td><td>Common during winter in cold climates.</td><td>16.0
15.5</td><td rowspan="2">Up to 16 V is possible when a battery charger is connected to the vehicle charging system.</td><td rowspan="2">Common service practice to recharge battery.</td></tr>
<tr><td rowspan="2">Normal driving and idling with typical electrical system loading.</td><td rowspan="2">Often.</td><td>15.0
14.5
14.0</td></tr>
<tr><td>13.5
13.0</td><td rowspan="2">Normal voltage range for a battery with a normal state of charge and typical electrical loading.</td><td rowspan="2">Often.</td></tr>
<tr><td rowspan="3">System voltage drops out of regulation when electrical load exceeds charging system capacity</td><td rowspan="2">Common when idling in gear with accessories in use.</td><td>12.5
12.0</td></tr>
<tr><td>11.5
11.0
10.5</td><td>Battery is in a low state of charge or is under heavy electrical loading. (Will still start engine).</td><td>Occasionally.</td></tr>
<tr><td>Rare. Extreme loading.</td><td>10.0
9.5
9.0
8.5</td><td rowspan="2">Battery is in a low state of charge and may have dead cells. (Will not start engine if below approx. 10.5 V).</td><td rowspan="2">Occasionally.</td></tr>
<tr><td>In engine stall or near–stall conditions system voltage may dip below 9 volts and then return to a steady state condition.</td><td>Very Rare.</td><td>8.0†
7.5
7.0
6.5
6.0</td></tr>
</table>

†Engines are usually not guaranteed to run below 8 V.
*Voltage at the component will be equal to or lower than the voltage at the battery depending on the amount of voltage drop within the wiring distribution system. Voltage ranges given are approximate (±0.5 V to ±0.75 V). Actual voltage will depend on individual vehicle design and will differ with normal production variations in charging system component characteristics.

M 2 TEMPERATURE CLASSIFICATIONS

The temperature parameters required for test are detailed in Table 9 and shall be defined in the PDS by the JLR Component Engineer in conjunction with the requirements in STJLR.18.233.

- **Operational Temperatures** define the minimum and maximum temperature the DUT can operate at for a prolonged period
- **Performance Operating Temperatures** (where applicable) define the maximum and minimum temperatures the DUT can operate at whilst meeting all of their functional performance targets

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- **Exposure Temperatures** define the minimum and maximum temperature the DUT can survive whilst in a non-operational state. This includes the assembly of the part, transport, storage and assembly into a JLR vehicle.
- **Maximum Soak Temperature** (where applicable) defines a short term maximum temperature for parts that must survive a solar or powertrain soak. The time the component shall withstand this temperature shall be defined in the PDS.

Table 9 - Test Temperatures

Tests	Temp	Description	Definition
Operational	T_{min}	Minimum Operating Temperature	-40°C
	T_{max}	Maximum Operating Temperature	As PDS
Performance	T_{Pmin}	Minimum Performance Operating Temperature	As PDS
	T_{Pmax}	Maximum Performance Operating Temperature	As PDS
Exposure	T_{Emin}	Minimum Exposure Temperature	-40°C
	T_{Emax}	Maximum Exposure Temperature	As PDS
Other	T_{soak}	Maximum Soak Temperature	As PDS
	T_N	Nominal Temperature	20 °C

M 3 WATER/FLUID INGRESS CLASSIFICATIONS

Water and fluid ingress classifications shall be defined in the component PDS and are determined in conjunction with STJLR.18.233 and STJLR.00.5056 as cascaded by the durability and robustness team.

M 4 SALT MIST TEST

Salt mist test classifications and durations shall be defined in the component PDS and are determined in conjunction with STJLR.18.233. Table 10 and Table 11 are provided for reference.

Table 10 - Salt Mist Test Durations

Classification	Duration
I	N/A
II	24
III	48
IV	168

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Table 11 - Salt Mist Test Classifications

	Component Mounting Location	Salt Mist Classification	Test Duration
A	**Vehicle Interior**		
A1	Upper cabin (including head restraints)	I	N/A
A2	Fascia top	I	N/A
A3	Rear parcel shelf	I	N/A
A4	Between Roof and Headlining	I	N/A
A5	Mirror pod (including windscreen mounted components)	I	N/A
A6	Upper tailgate pod	I	N/A
A7	Top of Door Casing	I	N/A
A8	Tailgate centre pod	I	N/A
A9	Mid cabin (between seat cushion and head restraint)	I	N/A
A10	Upper Fascia, Cluster	I	N/A
A11	Steering wheel and column	I	N/A
A12	Within Door Casing (Above Wade-Line/cabin side of Water Curtain)	I	N/A
A13a	Lower Cabin, below the splash line, as set out in the Interior Wet Zone definition (STJLR.00.5056)	II	24 Hrs
A13b	Lower Cabin, below the immersion line, as set out in the Interior Wet Zone Definition (STJLR.00.5056)	II	24 Hrs
A13c	Lower Cabin, not in the Splash or immersion zones, as set out in the Interior Wet Zone Definition (STJLR.00.5056, Section H)	II	24 Hrs
A14	Lower Fascia	I	N/A
A15	Within Door Casing (Below Wade-Line /cabin side of Water Curtain)	I	N/A
A16	Behind rear seats (between seat back and rear bulkhead)	I	N/A
B	**Engine Bay**		
B1	Front, Sides of Engine Bay, Bonnet underside	IV	168 Hrs (Note 1)
B2	Rear (behind engine)	IV	168 Hrs (Note 1)
B3	Plenum	IV	168 Hrs (Note 1)
B4	Forward of cooling pack	IV	168 Hrs (Note 1)
B5	On engine / transmission	IV	168 Hrs (Note 1)
B6	Headlamp zone (extends between B4 and B1)	IV	168 Hrs (Note 1)
C	**Load Space**		
C1	Underside of Boot Lid / Upper Tailgate (when opened & closed)	I	N/A
C2	Between Rear Parcel shelf and trim	I	N/A
C3	Boot side areas above the splash line, as set out in the Interior Wet Zone Definition (STJLR.00.5056)	I	N/A
C4	Under boot (internal)	I	N/A
C5	Lower Tailgate (for vehicles with split tailgate) on dry side of water curtain	I	N/A
C6	Rear lamp zone	I	N/A
D	**Vehicle Exterior**		
D1	On roof exterior	III	48 Hrs
D2	On bonnet exterior	III	48 Hrs
D3	On boot / tailgate exterior	III	48 Hrs
D4	Under body - rear (under boot floor, inner rear bumper)	IV	168 Hrs (Note 1)

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	Component Mounting Location	Salt Mist Classification	Test Duration
D5	Under Floor - sides	IV	168 Hrs (Note 1)
D6	Under Floor - tunnel	IV	168 Hrs (Note 1)
D7	Under Floor - within 150 mm of exhaust	IV	168 Hrs (Note 1)
D8	Between Wheel-Arch Liners and Body	IV	168 Hrs (Note 1)
D9	Door Exterior (handles, mirrors)	III	48 Hrs
D10	Door/Wing Exterior upright panel	III	48 Hrs
D11	Within Door Casing (Above Wade-Line/outside of Water Curtain)	III	48 Hrs
D12	Within Door Casing (Below Wade-Line/outside of Water Curtain)	III	48 Hrs
D13	In Load space area (including truck drop gate)	III	48 Hrs
E	**Wheels**		
E1	On hub / Near brake disc	IV	168 Hrs (Note 1)
E2	On moving suspension component	IV	168 Hrs (Note 1)
E3	On the wheel	IV	168 Hrs (Note 1)
E4	On wheel rim/ inside the Tyre	IV	168 Hrs (Note 1)
Note 1:	**Appropriate duration shall be selected based on component FMA, FMEA and/or component functional and appearance requirements with respect to corrosion resistance.**		

M 5 CHEMICAL SELECTION

The chemicals for test and application methods shall be defined in the component PDS in conjunction with the requirements in STJLR.18.233. Table 12 is provided for reference.

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Table 12 - Chemicals

Chemicals (Note 4)	Vehicle compartments/locations (Note 1)				Application Methods	Test Temperature	Test Duration
	Engine Bay	Exterior	Luggage Compartment	Passenger Compartment	(Note 2)	(Note 3)	hh:mm
Diesel Fuel	x				I,III,IV,V	T_6	22:00
"Bio" Diesel	x				I,III,IV,V	T_6	22:00
Unleaded Petrol/Gasoline	x				I,III,IV,V	TR	00:10
Methanol	x				II,III,IV,V,VI	TR	00:10
Engine Oil	x				II,III,IV,V	T_6	22:00
Differential Oil	x				II,III,IV,V,VI	T_6	22:00
Transmission Fluid	x				II,III,IV,V,VI	T_6	22:00
Hydraulic Fluid	x				II,III,IV,V	T_6	22:00
Greases	x				II,III	T_6	22:00
Silicone Oil	x				I,II,III,V	T_6	22:00
Cleaning Solvent	x				I,II,III	TR	00:10
Contact Spray	x				I,II,III	TR	22:00
Antifreeze Fluid	x				I,III,IV,V,VI	T_6	22:00
Calcium Chloride (Solution)	x	x			I, II, III	TR	22:00
Urea	x	x			II,III,V	T_6	22:00
Protective Lacquer	x	x			I,II	TR	22:00
Protective Lacquer Remover	x	x			I,III,IV,V	T_6	22:00
Runway De-icer	x	x			I,II,IV	TR	02:00
Vehicle Washing Chemicals	x	x			I,II,III,IV,V	TR	02:00
Window Screen Washer Fluid	x	x	x		II,III,IV,V	TR	02:00
Cold Cleaning Agent	x	x	x		I,II,III,IV,V,VI	TR	22:00
Brake Fluid	x		x		II,III,V	T_6	22:00
Battery Fluid (Note 5)	x			x	III,V	TR	22:00
Denatured Alcohol	x	x	x	x	I,II,III,IV,V	TR	00:10
Cavity Protection		x			II,III	TR	22:00
Damper Oil		x			I,II,III,IV,V	TR	22:00
Wheel Cleaner		x			I,II,III,IV	TR	02:00
Glass Cleaner		x	x	x	I,III	TR	02:00
Ammonia Containing Cleaner		x	x	x	II,III,V	TR	22:00
Leather Wax		x	x	x	II, III	TR	22:00
Kerosene			x		II,III,IV,V	TR	00:10
Deodorizer Spray			x	x	I	TR	22:00
Interior Cleaners			x	x	I,III	TR	02:00
Acetone				x	I,II,III	TR	00:10
Perspiration				x	II,III,V	TR	22:00
Cosmetic Products eg Creams/Moisturisers				x	II,III	TR	22:00
Refreshment containing Sugar & Caffeine				x	III,IV	TR	22:00
Cream, Coffee Whitener				x	III,IV	TR	22:00

Note 1: For other packaging locations, the chemicals to be tested, and the application methods shall be defined by the JLR component engineer and communicated to the supplier in the Product Design Specification.

Note 2: Application methods: I - Spray: II - Brush: III - Wipe: IV - Pour: V - Dip: VI - Immerse

Note 3: TR is defined as 23 °C ± 5 °C

Note 4: The listed chemicals and any additional chemicals shall be considered and included in the tests if an exposure risk is identified in the FMEA. Additional chemicals shall be considered by the JLR Component Engineer and included in the Product Design Specifications.

Note 5: Within a radius of 200 mm from the centre of the battery and below the top of the battery.

Note 6: Definitions of most of the chemicals can be found in ISO 16750-5 : 2011

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M 6 TEST PURPOSES AND TYPICAL ENVIRONMENTAL EFFECTS

The following table provides a test purpose and typical environmental effects of each test. Test Purpose and Typical Environmental Effects are included at the head of each test.

Tests are grouped into

- CL – Climatic Tests
- ME – Mechanical Tests
- L – Lifetime Tests
- IN – Ingress Tests

Table 13 - Summary of Tests and their purposes

Title	Test Purpose	TYPICAL ENVIRONMENTAL EFFECTS
Climatic Tests		
Low Temperature Exposure **(CL-LTE)**	To verify that the component can withstand the effects of extreme low temperatures encountered in vehicles used in cold climates. Additionally, cold temperatures that may be experienced in shipping the component.	Change in physical properties of materials resulting in: • Hardening and embrittlement • Binding or loosening due to contraction of dissimilar materials • Crack/crazing • Loss of lubrication and lubricant flow • Static fatigue • Condensation and freezing of water
Low Temperature Operation **(CL-LTO)**	To verify that the component operates properly at low temperatures encountered during vehicle cold weather operation.	All effects listed above plus. Change in physical properties of materials resulting in: • Changes in electronic components and performance of electronic circuits • Changes in performance of transformers and electro-mechanical components
High Temperature Exposure **(CL-HTE)**	To verify that the component can withstand the effects of high temperatures which occur in the vehicle as a result of high ambient temperatures combined with solar radiation and/or vehicle self-heating.	Changes in physical properties and/or dimensions of materials resulting in: • Binding due to expansion of dissimilar materials • Reduction in viscosity of lubricants. Outward flow of lubricants from seals • Distortion or deterioration of seals, gaskets etc. • Permanent setting of resinous materials such as plastics and rubber • High internal pressures created in sealed parts • Cracking or crazing of organic materials

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Title	Test Purpose	TYPICAL ENVIRONMENTAL EFFECTS
High Temperature Operation **(CL-HTO)**	To verify that the component operates properly at high temperatures which occur in the vehicle as a result of high ambient temperatures combined with solar radiation and/or vehicle self-heating.	All effects listed above plus: • Changes in the stability and/or performance characteristics of electronic circuits and components • Overheating of transformers, electro-mechanical components and power devices • Altering the operation/release margins of relays and magnetic or thermally activated devices
Temperature Step Test **(CL-TST)**	To verify the component is able to operate at all temperatures within its temperature specification	• Intermittent or erroneous operation
Powered Thermal Cycle **(CL-PTC)**	To verify that the component is able to operate in changing temperature conditions encountered in the vehicle environment.	• Breakage or cracking due to stresses created by expansion and contraction • Intermittent or erroneous operation
Thermal Shock in Air **(CL-TSA)**	To verify that the component can withstand the effects of rapid changes in ambient air temperature which occur in the vehicle environment and during shipping and handling.	• Binding or loosening of moving parts • Deformation, cracking or fracture of materials • Leaking of sealed compartments • Changes in physical alignment of related parts and optical equipment
Thermal Shock Endurance **(L-TSE)**	To verify that the component and its PCBA(s) are free from failure caused by thermally induced stresses which can occur during it's intended useful life in the vehicle. This test is intended specifically for assessing resistance to fatigue failures, particularly solder fatigue cracking, as listed in "Typical Environmental Effects." This test is N/A for components without a PCBA	Fatiguing of materials due to stresses created by contraction and expansion. Fatigue failures can occur in:- • Solder • Solder connections • PCB traces • Hybrid circuitry • Conformal Coating
Composite Temperature Humidity Test **(CL-THC)**	To verify there are no deficiencies in the design of the electrical / electronic module caused by "breathing", including effect of trapped water freezing leading to	Physical and chemical deterioration of material: • Swelling of hygroscopic materials

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Title	Test Purpose	TYPICAL ENVIRONMENTAL EFFECTS
	cracks and fissures. Additionally the effect of frosting	• Degradation in electrical and thermal properties in insulating materials • Electrical Shorts • Binding of moving parts • Oxidation and/or galvanic corrosion of metals • Degradation of electrical components/circuits. Degradation of image transmission through glass or plastic optical elements
Dewing Test **(CL-DEW)**	To reveal design deficiencies caused by condensation on electrical and electronic circuits within the module.	Electrical failures as a result of • Increased leakage currents • Electromigration (Dendritic Growth) and Electrical Shorts • Corrosion
Water/Fluids Ingress (IN-W)		
Ingress Tests (IPX1-IPX9K) **(IN-W)**	To verify the component can withstand the effects of the moisture exposure in the relevant specification	• Formation of electrochemical migration (Dendrites) which may cause malfunction or result in an unsafe condition • Corrosion due to direct exposure to water or the high humidity levels caused by the water • Damage to seals • Removal of required labels • Loss of lubricants between moving parts
Immersion with Thermal Shock **(IN-TSI)**	To verify the component can withstand the effects of complete immersion including sudden immersion in cold water when operating at a high temperature (Thermal Shock)	• Cracking or separation of bonded materials such as case and potting interface • Damage to seals • Formation of electrochemical migration (Dendrites) which may cause malfunction or result in an unsafe condition • Corrosion due to direct exposure to water or the high humidity levels caused by the water • Removal of required labels • Loss of lubricants between moving parts
Splash with Thermal Shock **(IN-TSS)**	To verify the component can withstand the effects of being splashed including being splashed with cold water when operating at a high temperature (Thermal Shock)	• Cracking or separation of bonded materials such as case and potting interface • Damage to seals • Formation of electrochemical migration (Dendrites) which may cause malfunction or result in an unsafe condition

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Title	Test Purpose	TYPICAL ENVIRONMENTAL EFFECTS
		• Corrosion due to direct exposure to water or the high humidity levels caused by the water • Removal of required labels • Loss of lubricants between moving parts
Dust		
Static test **(IN-D)** Blowing Dust Test **(IN-BD)**	To verify that the component can withstand the effects of the accumulation of fine dust particles resulting from continued operation of the vehicle in dusty driving conditions.	• Abrasion/erosion of surfaces • Penetration of seals • Degradation of electrical circuits caused by reduced heat dissipation capacity due to clogging, coverage, etc.. • Clogging of openings and filters • Physical interference of moving parts
Mechanical		
Powered Vibration Endurance - Method **(ME-V)**	To verify that the component or component/bracket assembly is capable of withstanding the effects of the vehicle vibration environment.	• Loss of wiring harness electrical connection. • Wire chafing • Intermittent electrical connections • Touching and shorting of electrical parts • Optical misalignment • Excessive electrical noise • Temporary loss of function due to acceleration forces on components (e.g. due to mass of a diaphragm-spring mechanism) • False operation of electro-mechanical components (relays etc.) • Loosening of particles and parts which have become lodged in circuits or mechanisms • Loosening of fasteners • Seal deformation • Component fatigue • Cracking/rupture • Excitation of tuned mass harmonic
Audible Noise - Vibration Method **(AN)**	To verify that the component or component will not emit audible noise (buzzes, squeaks, rattles, etc.) in the vehicle that could cause annoyance to the customer.	Buzzes, squeaks, rattles, etc. which may be annoying to vehicle driver or passengers.

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Mechanical Shock Tests **(ME-S)**		
Package Drop	To verify that the component shipping container provides sufficient protection from mechanical impact shocks resulting from being dropped during shipping and handling.	• Cracking or breaking of enclosures, mounting points, connectors and mechanical interfaces • Breaking of parts within the component assembly • Damage of electrical connections and continuity within the component enclosure • Malfunction of electro-mechanical devices, mechanisms and controls contained within the component • Loosening of covers and seals • Cracking of substrates
Handling Drop	To verify that the component is not damaged by mechanical impact resulting from being dropped onto hard surfaces from distances up to 76 cm and is capable of withstanding mechanical shock impulses encountered during vehicle operation, shipment, towing, etc..	
Medium Mechanical Shock	To verify that the component is not damaged by short drops (up to 20 cm) onto conveyors, benches, etc. during normal assembly, service, vehicle installation, shipping, etc. and is capable of withstanding mechanical shock impulses encountered during vehicle operation, shipment, towing, etc..	
Low Mechanical Shock	To verify that the component is not damaged by short drops (up to 10 cm) onto conveyors, benches, etc. during normal assembly, service, vehicle installation, shipping, etc. and is capable of withstanding mechanical shock impulses encountered during vehicle operation, shipment, towing, etc..	
High Repetition Mechanical Shock	To verify that the component is not damaged by repetitive shock impulses as measured for the specific mounting location on the vehicle	
Connector and Lead/Lock Strength **(ME-C)**		
Connector Push Test	To verify that the component-side electrical connector, leads and locking mechanisms are of sufficient strength to withstand mechanical stresses (pulling, twisting, etc.) expected in the	• Loss of connection • Intermittent connection
Connector Torque Test		
Connector Durability Test		

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Lead/Lock Pull Test	vehicle and during vehicle assembly, service and repair.	
Other Climatic Tests		
Salt Mist Atmosphere **(CL-SALT)**	To verify that the component can withstand the effects of continued exposure to salt water and/or salt atmosphere conditions resulting from vehicle operation on salted roadways and/or in high salt-humidity atmosphere regions.	• Impairment due to salt deposits • Production of conductive coatings • Corrosion due to electro-chemical reaction • Clogging or binding of moving parts
Solar Radiation **(CL-SOL)**	To verify components mounted in areas exposed to the sun can withstand prolonged periods of solar radiation	• Degradation of materials (cracking, softening, warping, embrittlement etc.) • Damage to optical sensors
Chemical Tests		
Corrosive Gases **(CL-CG)**	To verify the component can withstand the effects of common atmospheric pollutant that can occur in higher concentrations in built up areas	
Chemical Resistance **(CR)**	To verify that the component can withstand the effects of chemicals (cleaning compounds, lubricants, automotive fluids, etc.) encountered during component manufacture, vehicle assembly, service and maintenance, and during vehicle use.	• Degradation of materials (cracking, softening, warping, etc.) • Deterioration of label adhesives • Discolouring of materials • Smudging of inks and dyes
Life Time Tests		
Controls Durability **(L-CD)**	To demonstrate that component driver/passenger operated controls have no wear out mechanisms which can occur during the intended useful life of the component in the vehicle. This test is N/A for components that do not have customer facing controls	• Contact wear • Fatigue of plastic and metal parts • Electrical intermittents/noise • Mechanical wear • Binding or sticking

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High Temperature Endurance **(L-HTE)**	To demonstrate that the component electronics are free from time-temperature dependent failure during the intended useful life of the component in it's vehicle application.	• Dielectric breakdown • Damage to Electrolytic Capacitors, Flash • memory or other components subject high temperature wear • Metal migration • Intermetallic growth • Electro-migration • Ionic contamination
Mechanical Wear-out **(L-MW)**	To demonstrate that mechanical or electro-mechanical (moving) parts contained within the component have no wear-out mechanisms which occur during the intended useful life of the component in the vehicle.	• Contact wear • Fatigue of plastic and metal parts • Audible noise • Mechanical wear • Binding or irregular operation
85/85 High Temperature - High Humidity Endurance **(L-THB)**	To demonstrate that the component electronics are free from high temperature-humidity dependent failure during the intended useful life in the vehicle application.	• Swelling of hydroscopic materials • Electrolytic corrosion of die metallization • Degradation in electronic materials/circuit performance • Current leakage • Dendritic growth

M 7 TNI/NTF Return Evaluation Process

GOOD:

1) Start with the normal return checking process, EOL or lab ambient temp/voltage evaluation.

2) Warranty analysis/CQIS:

Pull VIN history to see if the customer returned for additional service.

Look at work ticket to see if the customer complaint and/or mechanic comments make sense for the removal of the module. Also look at what other parts were replaced with this one that may have been the real failed part. If part appears to have been wrongly pulled review Service Procedures in Service Manual with the JLR Component Engineer and consider a possible update to help field techs with future repairs.

You may also consider asking the FCSD Warranty Commodity Engineer to assist with a charge back.

Create a log to track service mistakes and list the detail for each returned part.

If the module appears to be suspect:

Look at customer complaint and D/PFMEA and attempt to determine what part of circuit/software could have caused the failure (possibly intermittent) and retest using the Performance Evaluation test with this knowledge as a focus for the evaluation. If failure is not replicated proceed to 'better' below or stop if so directed by the JLR Component Engineer.

BETTER:

All of the above and:

Monitored Temp/Power cycled module for 24 hours.

Powered/monitored vibration test, three sweeps per axis.

Monitored Power/Humidity test for two cycles.

If failure is not replicated proceed to 'best' below or stop if so directed by the JLR Component Engineer.

BEST:

All of the above and:

With the assistance of the JLR Warranty Commodity Engineer return module to a vehicle for a few days and have driver monitor the suspected function to access proper operation and look to replicate the customer's original complaint. If failure is not replicated run the Performance Evaluation and if pass stop.

Suggestion: if modules pass all of the above request permission from JLR Engineering to save them and start a part recycle program if you receive a large enough number of TNI returns and use as service stock and track field performance of TNI returns.

With your JLR Component Engineer you may also want to look at other modules/parts in the vehicle system that could have caused the customer complaint e.g. loss of signal or ground, intermittents, software interaction etc..

N TERMINOLOGY AND DEFINITIONS

N 1 Accelerated Test.

A test which accelerates the wear or deterioration of an item by subjecting it to environmental stresses and/or loading conditions beyond those normally encountered in it's application and/or by increasing the frequency with which loads or stresses are applied during normal use. Accelerated tests are typically conducted at or above maximum stress or loading conditions for which the item is rated. Accelerated test methods are useful in attribute testing, life testing, reliability testing and development testing.

N 2 Accelerated Life Test.

A test specifically designed and conducted with the intent of obtaining failure rate, time-to-failure and/or cycles-to-failure data or to obtain information useful in determining or understanding failure modes and mechanisms.

N 3 Attribute Test.

A test which verifies that an item has a specific attribute or set of attributes (e.g. operates as specified at -30 °C, has 1,000 hours life at max. rated temperature). Qualification, Certification, Design Verification, Production Validation, In-Process, Demonstrated Life and Reliability Demonstration tests are all examples of attribute tests. Attribute tests are usually specified as a "requirement" on engineering drawings, specifications and/or contractual agreements and thus form the basis for establishing design requirements.

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N 4 Audit Test.

A test performed on a sample of items (incoming material, manufactured parts, process equipment, etc.) to verify compliance with a standard, practice, procedure or specification.

N 5 Burn-in.

A test, usually conducted on production parts at the manufacturing facility, which is used for the purpose of screening out substandard (weak) product, auditing quality levels and/or to help gain information useful in improving quality levels. Typically, in a Burn-in test, production parts are operated continuously at maximum or above maximum rated power or voltage at ambient temperature or maximum rated temperature. The test duration is usually between 8 and 48 hours depending on the nature of product failure mechanisms, test acceleration factors and desired quality levels.

N 6 Climatic Test Chamber.

An enclosure or space in which the temperature/humidity condition specified in this TPJLR can be achieved.

N 7 Component Engineer.

The Jaguar Land Rover Engineer responsible for the DUT.

N 8 Demonstrated Life Test.

A life test method which specifies a predetermined number of hours and/or cycles as a bogie or “pass” criteria. Demonstrated Life Test “pass” criteria is established through correlation to vehicle environmental conditions and 10 year/150,000 mile duty cycles or according to requirements or recommended practice defined in corporate or industry standards.

N 9 Design Verification (DV).

The act of determining the vehicle environment (temperature, vibration, duty cycles, loads, etc.) and defining and conducting all tests and evaluations necessary to assure the adequacy of a design for the vehicle environment.

N 10 Design Verification Plan (DVP).

A test plan which lists all tests and evaluations, acceptance criteria, timing and sample size requirements for Design Verification of a specific product design or design level.

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N 11 Design Verification Report (DVR).

A report documenting Design Verification test results and actual test completion dates.

a Design Verification Plan & Report (DVP&R).

b “Design Verification Plan and Report,” used for documenting Design Verification plans and reports.

c A design verification plan and report for a specific product or product design level.

d Design Verification (DV) Test.

e Any test specified in a Product Verification Specification (PV Spec) or Design Verification Plan & Report (DVP&R) which is intended for use in verifying a design attribute.

f A non-standard test procedure (ie. not published in an authoritative document such as ES, TPJLR, SAE, MIL-STD, IEC, ASTM, etc.), which is documented and specified in a PV Spec and/or DVP&R.

N 12 Development Testing.

Informal testing performed for developmental purposes during the development stages of a program, typically to gain insight into product design attributes such as operating limits, robustness to environmental conditions, etc.. Examples of development testing include step-stress testing, characterization testing etc.. Also see Pre-DV testing.

N 13 Device Under Test (DUT).

A DUT is a module/component to be tested in one or more of the tests in this document.

N 14 Energy Dispersive X-Ray Spectroscopy (EDX).

A technique using X-rays used to perform elemental analysis on a test subject, relying on a the effect of an X-ray Source exciting a sample

N 15 Engineering Specification (ES).

A formally released document which, along with associated engineering drawings, operationally defines product design requirements in terms appropriate for use by the manufacturing facility. The Engineering Specification contains all Production Validation (PV), In-Process (IP) and Functional Tests, acceptance criteria and any other items not appropriate or practical to include in an engineering drawing.

N 16 Environmental Stress Screen (ESS).

A test, usually performed on 100 % of production samples, which utilizes environmental stresses to stimulate relevant failures for the purpose of screening out substandard (weak) parts, assemblies or sub-assemblies.

N 17 Finite Element Analysis (FEA).

A mathematical technique for analysing stress where a physical structure is split into substructures called “Finite Elements”

N 18 Functional Check.

An abbreviated Functional Test or Performance Evaluation, usually conducted at room or ambient temperature only, which shows whether or how well a product functions.

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N 19 Functional Evaluation.

A test which evaluates the functionality of a product at worst case operating conditions (usually temperature and supply voltage). Functional Evaluations are specified for products which are required to perform intended functions under extreme conditions outside the range (usually temperature and supply voltage) of guaranteed performance.

N 20 Functional Test.

A test for a product which shows whether or how well it performs a function or set of functions. A Functional Test is typically comprised of a Performance Evaluation and may also include a Functional Evaluation. Functional Tests are specified in the product Engineering Specification (ES).

N 21 Highly Accelerated Life Test (HALT)

This is an evaluation method to determine the operational and destruct limits of an electrical/electronic design/processed component using special chambers. The environments include but are not limited to min/max temperature, vibration and thermal shock

N 22 Highly Accelerated Stress Screen (HASS)

This is a process monitor or audit method that uses the results of the HALT tests to create a screen which can be used to verify minimum operational limits for the environments listed above.

N 23 In Process (IP) Testing.

The act of assuring that production parts continue to meet design intent through periodic tests and evaluations.

N 24 In Process (IP) Test.

A test specified in a Product Design Specification (PDS), Product Verification Specification (PV Spec) and/or Control Plan which is intended for use in an IP testing activity.

N 25 Infant Mortality.

a A term referring to product failures which typically occur early in the product life cycle and may be attributed to substandard (defective) parts and/or manufacturing processes.

b The left-hand portion of the classic “Bathtub” curve where the overall failure rate for a population of parts is decreasing rapidly with time.

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N 26 Key Life Test.

An Accelerated Life Test or Demonstrated Life Test which is designed to evaluate a product's robustness to one or more "key" (or relevant) wear out modes or mechanisms. By definition, Key Life Test acceleration factors must be correlated to actual vehicle environmental conditions and duty cycles for 90th and/or 95th percentile customer usage for 10 year/150,000 miles useful life. Acceptance or "pass" criteria for 10 year/150,000 mile usage must be established. Since most products have more than one relevant wear out mode or mechanism, several Key Life Tests may be required to evaluate a given product. Key Life Tests should be specified as appropriate in Product Verification Specifications, Design Verification Plans and Engineering Specifications.

N 27 Life Test.

A test which may be used to help predict the expected life of a product in it's intended vehicle application(s). Typically, Life Tests are conducted to characterize product wear-out. In order to provide acceptable results, the test should yield time-to-failure or cycles-to-failure data sufficient for making calculations to desired confidence levels.

N 28 Limit Test.

A test which is used to determine the operating limits or destruct points of a product. Limit Tests typically employ step-stress techniques (eg. increasing temperature, voltage, or mechanical load in increments) until the product becomes non-functional and/or a catastrophic failure occurs.

N 29 Mission Life.

The expected operational life of the component in a vehicle.

N 30 Module Conditioning.

A term referring to a production test performed on all or a sample of production parts (electronic modules) for the purpose of screening out substandard (weak) product, audit quality levels and/or to help gain information useful in improving quality levels. In contrast to Burn-in or ESS, Module Conditioning is characterized by temperature cycling between minimum and maximum rated operating temperature and cycling of module inputs and outputs.

N 31 Performance Evaluation.

A test which measures a product over it's range (usually temperature and supply voltage) of guaranteed performance.

N 32 Pre-DV Testing.

A term referring to informal product development testing performed to ensure that a product design meets design requirements and will pass Design Verification testing. Also see Development Testing.

N 33 Printed Circuit Board Assembly (PCBA).

A PCB that has been populated with components

N 34 Primary Side (of a PCB).

That side of a printed circuit board (PCB) that is so defined on the master drawing (it is usually the side that contains the majority of components). This side is sometimes referred to as the component side (as defined in IPC-T-50).

N 35 Product Design Specification (PDS).

The full specification of the component. This specification should be in sufficient detail that, if supplied to 2 different suppliers will result in designs that would be interchangeable in the field.

N 36 PV.

A JLR acronym with multiple meanings:

a Product Verification Specification (as in PV Spec).

b Production Validation (as in PV testing).

c Process Verification (as for PV Plan). It is strongly recommended that the use of acronym “PV” by itself is avoided and that the full term be used.

N 37 Process Verification.

The act of assessing the capability of a specific manufacturing process, process step or operation for a specific product and/or product design.

NOTE: Process Verification should not be confused with Production Validation (PV) testing or Product Verification (PV) Specification.

N 38 Process Verification Plan (PVP).

A comprehensive listing of Process Verification tests, test timing and sample size requirements necessary for a process verification activity.

N 39 Process Verification Report (PVR).

A report documenting the results of a specific process verification program.

N 40 Process Verification Test.

A test, measurement or evaluation specified in a Process Verification Plan which is intended to measure the capability of specific machine, operation or manufacturing process step.

N 41 Product Verification Specification (PV Spec).

a A Product Verification Specification (PV Spec) completed for a specific vehicle system, subsystem, assembly or component.

b A document which lists all engineering evaluations and tests required to establish the vehicle environment, assure that the design is adequate for the vehicle environment, and assure that production parts meet design intent. PV Specs are “master” specifications that are carried over from year to year. They are continually updated and refined with state-of-the-art test techniques and to accommodate major changes in product concepts or engineering requirements. The PV Spec is the primary reference in preparing Design Verification Plans (DVP’s) and in specifying Production Validation (PV) tests and In Process (IP) tests contained in product ESs and Control Plans.

NOTE: The term PV Spec should not to be confused with the term Production Validation (PV) testing or the term Process Verification.

N 42 Production Validation (PV).

The act of assuring that initial production parts made from production tools and via production processes will meet design intent through appropriate tests and evaluations.

NOTE: Production Validation (PV) should not to be confused with the terms Product Verification Specification (PV Spec) or Process Verification.

N 43 Production Validation (PV) Test.

A test specified in an Engineering Specification (ES) and/or a Product Verification Specification which is intended to be used in Production Validation programs for a specific product.

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N 44 Production Validation (PV) Testing.

Testing performed to fulfil the requirements of a Production Validation program for a specific product as specified in a Production Validation Plan and the released product Engineering Specification (ES).

N 45 Relative Humidity (RH).

Ratio of the amount of water vapour present in air expressed as a percentage of the amount needed for saturation at the same temperature.

N 46 Reliability Demonstration Testing.

Testing performed to demonstrate that a product meets reliability requirements specified in a contractual document. Such reliability requirements must be stated in measurable terms and must explicitly define all relevant evaluation and test methods, sample sizes and statistical acceptance criteria.

N 47 Reliability Growth Testing.

Reliability testing performed during the development stages of a program for the purpose of quantitatively assessing and improving product reliability. Reliability growth testing is, by nature, a test-analyze-and-fix process.

N 48 Reliability Testing.

Testing performed to assess, in quantitative terms, the reliability of a product, typically in MTTF (Mean Time to Failure), failure rate (in failures per million hours or failures per million cycles, etc.) or B10 life.

N 49 Scanning Electron Microscopy (SEM).

An imaging technique that relies on a focused beam of electronics being scanned over the surface of a sample. This process provides information about the surface topography and composition. SEM

N 50 Standard atmospheric conditions for measurements and tests.

As defined in BS EN 60068-1:1995 for the condition where air temperature lies in the range (15 to 35) °C, relative humidity (25 to 75) % and air pressure (86 to 106) kPa.

N 51 Step-Stress Testing.

Testing in which the stress or stresses applied to the item under test are increased in incremental steps. Step-Stress techniques are usually used in performing Limit Testing and in determining acceleration curves required for accelerated testing.

N 52 Surrogate data.

Data and/or test results from a previous successful validation test that will be used as a substitute for a DV or PV test on a similar module/assembly which has a minor change or changes.

Examples:

An instrument cluster has the same electronics as the previous model year design but the case/mounting has changed. The test data for the electronic validation will be used as surrogate, the only testing required will be for the mechanical verification.

The manufacturing line/process that will be used to build a module with minor electronic changes was used to build the current model year module. The validation data for the previous model year will be used as surrogate and the new module will require a WCCA for the circuit changes, verification that there are no new pad geometries. When both of these are confirmed the PV testing will require samples from the PSW run go through the Performance Evaluation test L 1.

N 53 Test-to-Failure.

The act of testing an item until it fails. Test-to-Failure is employed in Limit Testing and all forms of non-accelerated and accelerated life testing in order to obtain quantitative data regarding when, why and/or under what conditions an item fails.

N 54 Thermal Cycle (JLR definition).

This is a temperature cycling test where the rate of change of the chamber temperature is equal to or less than 5 °C/min.

N 55 Thermal Shock (JLR definition).

This is a temperature cycling test where the rate of change of the component temperature is greater than or equal to 10 °C/min. The monitor point shall be the one identified in section H 1.5 f.

N 56 Things Gone Wrong (TGW)

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N 57 Worst Case Circuit Analysis (WCCA).

(Extreme Value Analysis) is a rigorous mathematical evaluation of circuit performance attributes against performance tolerance limits under simultaneous existence of all the most unfavourable conditions being at realizable limits.

WCCA shall include the following:

- Circuit description
- System schematic
- Bill of Materials
- Theory of operation
- Description of functionality provided by each component
- Specifications
- Environmental conditions (temp, voltage, EMC, ground shifts, interface)
- Component specifications and assumptions
- Analysis schematic (with and without parasitics)
- Worst case analysis
- Stress analysis
- A/C analysis (if required)
- Transient analysis
- Summary
- Bench test results

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APPENDIX A – INDEX OF FIGURES AND TABLES

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Z CHANGE HISTORY

Issue	Date	Description of Change
1	29 Jun 2015	First issue
2	01 Dec 2015	Corrected Section L 4 sub-groups (PP 48 & 49). G 3 references throughout document corrected to G 1.3. K 7.3 d. wording corrected.
3	25 May 2018	Corrected G 1.4 to G 1.7.2, H 2.1, K 1, K 10.4.4, K 15 for text errors. L 1.6 Flow Chart re-drawn, Table L 4 sequence table updated to remove unused sequences. Table L 9 corrected. Many document format changes to increase readability. Tables L 2 and L 3 combined and changes to text. L 2 onwards adjusted to account for removal of L 3 table.
4	8 Feb 2021	• Replacement of the Condensing and Non Condensing tests • Addition of Ice Water Immersion and Splash tests • Modification of the Audible noise test • Addition of new ingress tests based on ISO 20653 • Rewritten Salt mist test classification and moved table. • Modifications to the acceptance tests • Addition of Temperature Step Test • Modifications to Powered Thermal Cycle • Addition of Corrosive Gas Test, and Solar Loading tests • Test groups and sequences now in STJLR.18.233 • Modifications to Vibration test methods and profiles • Removal of supplier deliverables to STJLR.18.233 • Modifications and additions to the definitions section • Numerous small changes to terminology • Reorganisation of headings. • Extensive cross references and links added
5	24th Jan 2022	Added additional information regarding accreditation requirements Contributions from Carmelo Malara (cmalara), Rishikesh Kumar (rkumar39), Shrikrishna

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